

A Study in Algebraic Number Theory

A report submitted in partial fulfillment of the requirements for the degree of

MASTER OF SCIENCE

by

DIBENDU PAL

Roll no.: **17MS098**

To the

Department of Mathematics and Statistics



**INDIAN INSTITUTE OF SCIENCE
EDUCATION AND RESEARCH, KOLKATA**

Supervised by

DR.SWARNENDU DATTA

Assistant Professor, Department Of Mathematics and Statistics, IISER Kolkata

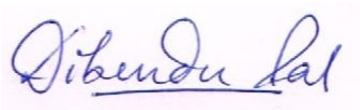
Abstract

This Report is a study to establish Class Number Formula for Quadratic fields which is an abelian extension and also study some properties of p -adic numbers from definition along with valuations. We first study generalised Dedekind Zeta Function on a number ring and then establish formula for number of ideal classes. We then move on to some prerequisites for studying p -adic numbers and then study them. Finally then we move to Valuations ending with Ostrowski Theorem.

Declaration

I, Mr. Dibendu Pal, Roll No. 17MS098, a student of IISER's 5 year integrated BS-MS Dual Degree Programme in Mathematics and Statistics Kolkata, hereby declare that the report titled "A Study in Algebraic Number Theory" is my own work, and it does not, to the best of my knowledge contains any materials written by anyone else, nor has it been submitted for any degree/diploma or any other academic award previously, unless due references are provided.

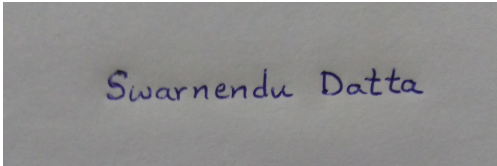
Dibendu Pal



Department of Mathematics and Statistics
Indian Institute of Science Education and Research, Kolkata
Mohanpur 741246, West Bengal, India.

Certificate from the Supervisor

This is to certify that the report entitled “A Study in Algebraic Number Theory” submitted by Mr. Dibendu Pal, Roll No. 17MS098, a student of Department of Mathematics and Statistics of the 5-year integrated BS-MS Dual Degree Programme of IISER Kolkata, is based upon his own work under my supervision. This is also to certify that neither the thesis nor any part of it has been submitted for any degree/diploma or any other academic award anywhere before, except where due references have been made. In my opinion, the thesis fulfills the requirements for the award of the degree of Master of Science.

A rectangular box containing a handwritten signature in blue ink that reads "Swarnendu Datta".

Dr. Swarnendu Datta

Assistant Professor

Department of Mathematics and Statistics

Indian Institute of Science Education and Research, Kolkata

Mohanpur 741246, West Bengal, India.

*To my mother and father
for their endless
love, support and encouragement.*

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Chapter 1

Introduction

The ancient topic of number theory is fundamentally concerned with arithmetic of integers. Algebraic number theory is distinguished from other branches of number theory by its application of abstract algebra techniques to number-theoretic problems. As a result, it is frequently regarded as a branch of algebra. The study of the finite field extensions of $\mathbb{Q}$, known as number fields, and their rings of integers, analogous to $\mathbb{Z}$, is the central topic of algebraic number theory.

The ring of integers $\mathcal{O}$ of a number field K is the subring of K consisting of all roots of all monic polynomials in $\mathbb{Z}[x]$. Unlike $\mathbb{Z}$, not all integer rings are UFDs, as one sees for instance by considering the factorization of 6 in the ring $\mathbb{Z}[\sqrt{-5}]$. However, they are what are known as Dedekind domains, which have the particularly nice property that every nonzero ideal factors uniquely as a product of nonzero prime ideals, which are all in fact maximal. In essence, prime ideals play the role in $\mathcal{O}$ that prime numbers do in $\mathbb{Z}$.

Let a prime ideal $\mathfrak{p}$ be in the integer ring $\mathcal{O}$ of a number field K , one can define a metric on K that measures the highest power of $\mathfrak{p}$ dividing the difference of two points in K . If the finite field $\mathcal{O}/\mathfrak{p}$ has characteristic p , then the completion $K_{\mathfrak{p}}$ of K with respect to this metric is known as a p -adic field, and the subring $\mathcal{O}_{\mathfrak{p}}$ that is the completion of $\mathcal{O}$ is called its valuation ring. In the case that $K = \mathbb{Q}$, one gets the p -adic numbers $\mathbb{Q}_p$ and p -adic integers $\mathbb{Z}_p$. The archimedean fields $\mathbb{R}$ and $\mathbb{C}$ are also completions of number fields with respect to the more familiar Euclidean metrics, and are in that sense similar to p -adic fields, but the geometry of p -adic fields is entirely different. For instance, a sequence of integers converges to 0 in $\mathbb{Z}_p$ if and only if it is eventually congruent to zero modulo arbitrarily high powers of p .

It is often easier to work with p -adic fields, as solutions to polynomial equations (Dihantine

equations) can be found in them by successive approximation modulo increasing powers of a prime ideal. The "Hasse principle" asserts that the existence of a solution to polynomial equations in a number field should be equivalent to the existence of a solution in every completion of it. (The Hasse principle does not actually hold in such generality, which partially explains the terminology.)

p-adic numbers were introduced in 1904 by the German mathematician K Hensel. They are used intensively in number theory. p-adic analysis was developed (mainly for needs of number theory) in many directions. This report is a culmination of a number of really good study materials on Algebraic Number Theory¹⁴⁵⁷² and p-Adic Analysis³⁶

Chapter 2

Ideals Distribution in Number Ring

Let the number field be K and the corresponding number ring $R = \mathbb{A} \cap K$. For each positive real t , we denote the no. of ideals with $||I|| \leq t$ as $i(t)$ and in the ideal class C with $||I|| \leq t$ as $i_C(t)$. Thus $i(t) = \sum_C i_C(t)$ which is finite. Let $n = [K : \mathbb{Q}]$.

(1.1) Theorem. There exists a real constant κ , only depending on R , s.t

$$i_C(t) = \kappa t + \epsilon_C(t)$$

where the error term $\epsilon_C(t)$ is $O(t^{1-1/n})$. In simpler words,

$$\left| \frac{\epsilon_C(t)}{t^{1-1/n}} \right| \leq M$$

for some $M \geq 0$ and $\forall t \geq t_0$ for some t_0

(1.2) Corollary. $i(t) = h\kappa t + \epsilon(t)$ where h is the ideal classes number in R and $\epsilon(t)$ is $O(t^{1-1/n})$

Sketch of the Proof . The theorem gives us that $\kappa = \lim_{t \rightarrow \infty} \frac{i_C(t)}{t}$. Calculating $i_C(t)$ is equivalent to counting the no. of elements in an ideal, and so there is a 1-1 correspondence:

$$\{I \in C; ||I|| \leq t\} \longleftrightarrow \{\alpha \in J; ||(\alpha)|| \leq t||J||\}$$

We construct a subset of R such that no two elements are associates and every non-zero element in R has a unit multiple. Such a set is the coset representatives of the group of units U of $R \setminus \{0\}$, and we count the elements of J in that set. By Dirichlet's Unit Theorem, $U = W \times V$ where W is the cyclic group of roots of unity in K and V is the free abelian subgroup of rank $r + s - 1$. Fixing V , we look at the map

$$V \subset U \subset R \setminus \{0\} \hookrightarrow \Gamma_R \setminus \{0\} \rightarrow \mathbb{R}^{r+s}$$

Here U maps onto $\Gamma_U \subseteq H \subseteq \mathbb{R}^{r+s}$ and H is hyperplane of $y_1 + \dots + y_{r+s} = 0$. Kernel of $U \rightarrow \Gamma_U$ is W and the restriction $V \rightarrow \Gamma_U$ is isomorphism.

Replace $\mathbb{R}^n \setminus \{0\}$ by $(\mathbb{R}^*)^r \times (\mathbb{C}^*)^s$. By improving the maps we get

$$V \subset U \subset R \setminus \{0\} \hookrightarrow (\mathbb{R}^*)^r \times (\mathbb{C}^*)^s \xrightarrow{\log} \mathbb{R}^{r+s}$$

The first map takes $\alpha \in \mathbb{R} \setminus \{0\} \mapsto (\sigma_1(\alpha), \dots, \sigma_r(\alpha), \gamma_1(\alpha), \dots, \gamma_s(\alpha))$

And $\log(x_1, \dots, x_r, z_1, \dots, z_s) = (\log|x_1|, \dots, 2\log|z_1|, \dots)$.

The coset Representatives for V we can get from a set of coset representatives for V' in group $(\mathbb{R}^*)^r \times (\mathbb{C}^*)^s$ which is called fundamental domain for V' say D .

Define $N(x_1, \dots, x_r, z_1, \dots, z_s) = x_1 \cdots x_r |z_1|^2 \cdots |z_s|^2$ in $(\mathbb{R}^*)^r \times (\mathbb{C}^*)^s$.

This $|N(x)| = ||(\alpha)||$. We want:

- To find D for V' in $(\mathbb{R}^*)^r \times (\mathbb{C}^*)^s$
- Counting elements $x \in \Gamma_J \cap D$ with $|N(x)| \leq t||J||$

So, no. of $x = |W| \cdot i_C(t)$

Lemma 1: Let $f : G \rightarrow G'$ be a homomorphism of abelian groups. Take S be a subgroup of G which is carried isomorphically onto $S' \subseteq G'$. Assume D' is a set of coset representatives for S' in G' . Then the total inverse image $D = f^{-1}(D')$ is a set of coset representatives for S in G .⁴

Applying lemma 1 to log function, V' maps onto Γ_U . So need a set D' of coset representatives for Γ_U in $\mathbb{R}^{r+s}$. Its total inverse image is D .

In general fix fundamental parallelepiped F for Γ_U in H and fix vector v passing through O

and orthogonal to H say $v = (1, \dots, 1, 2, \dots, 2)$. Then

$$D = \{x \in (\mathbb{R}^*)^r \times (\mathbb{C}^*)^s : \log x \in F \oplus \mathbb{R}v (= D')\}$$

Here $D = aD$ for all $a \in \mathbb{R}, a \neq 0$. Define $D_a = \{x \in D : |N(x)| \leq a\}$.

Using this we get $D_a = \sqrt[n]{a} D_1$. So we only need to count points in $\Gamma_J \cap \sqrt[n]{t} \|J\| D_1$

Theorem 2.: Let Γ be an n -dimensional lattice in $\mathbb{R}^n$ and let B be a bounded subset of $\mathbb{R}^n$ with a "sufficiently nice" boundary. Then, for $a \in \mathbb{R}$,

$$|\Gamma \cap aB| = \frac{\text{Vol}(B)}{\text{Vol}(\Gamma)} a^n + \gamma(a),$$

where $\gamma(a) = O(a^{n-1})$.

Here "sufficiently nice" boundary means $(n-1)$ Lipschitz parametrizable i.e boundary is covered by images of finitely many Lipschitz functions $f : [0, 1]^{n-1} \rightarrow \mathbb{R}^n$. Applying the theorem to Γ_J as D_1 has such boundary and using $\text{Vol}(\Gamma_J) = \text{Vol}(\Gamma_R) \|J\|$,

$$|\Gamma_J \cap \sqrt[n]{t} \|J\| D_1| = \frac{\text{Vol}(D_1)}{\text{Vol}(\Gamma_R)} t + O(t^{1-1/n})$$

Hence, $i_C(t) = \kappa t + O(t^{1-1/n})$ where

$$\kappa = \frac{\text{vol}(D_1)}{|W| \cdot \text{vol}(\Gamma_R)} = \frac{2^s \text{vol}(D_1)}{|W| \sqrt{|\text{disc}(R)|}}$$

□

Now let D_1^+ consists of all points of D_1 s.t $x_1, \dots, x_r \geq 0$. Then $\text{vol}(D_1) = 2^r \text{vol}(D_1^+)$.

Let the fundamental parallelepiped F has form $\sum_{k=1}^{r+s-1} t_k v_k : 0 \leq t_k \leq 1$ for some $\mathbb{Z}$ basis $\{v_1, \dots, v_{r+s-1}\}$ for Γ_U .

For each k , set $v_k = (v_k^{(1)}, \dots, v_k^{(r+s)})$. Now a point $(x_1, \dots, x_r, z_1, \dots, z_s)$ in D_1^+ is character-

ized by:

$$\begin{aligned} \log(x_1) &= \sum_{k=1}^{r+s-1} t_k v_k^{(1)} + u \dots \log(x_r) = \sum_{k=1}^{r+s-1} t_k v_k^{(r)} + u \\ 2\log|z_1| &= \sum_{k=1}^{r+s-1} t_k v_k^{(r+1)} + 2u \dots 2\log|z_s| = \sum_{k=1}^{r+s-1} t_k v_k^{(r+s)} + 2u \end{aligned}$$

Now any element $(x_1, \dots, x_r, z_1, \dots, z_s)$ of D_1^+ has $|N(x)| \leq 1$ and if the log of first $r+s-1$ belongs to F then

$$\log(x_1), \dots, \log(z_{s-1}) \geq 0 \implies |x_1|, \dots, |x_r|, |z_1|^2, \dots, |z_{s-1}|^2 \geq 1 \implies |z_s| \leq 1 \implies u \leq 0$$

Set $t_{r+s} = e^u$ and introduce polar coordinates (ρ_j, θ_j) for each z_j we get D_1^+ as $(x_1, \dots, x_r, \rho_1 e^{i\theta_1}, \dots, \rho_s e^{i\theta_s})$ satisfying following equations

$$x_j = t_{r+s} e^{\sum_{k=1}^{r+s-1} t_k v_k^j}, \quad \rho_j = t_{r+s} e^{\frac{1}{2} \sum_{k=1}^{r+s-1} t_k v_k^{r+j}}, \quad \theta_j = 2\pi t_{r+s+j}$$

With $t_{r+s} \in (0, 1]$ and all other $t_k \in [0, 1]$.

Now to get formula for κ

$$\begin{aligned} \text{vol}(D_1^+) &= \int_{D_1^+} \rho_1 \cdots \rho_s dx_1 \cdots dx_r d\rho_1 \cdots d\rho_s d\theta_1 \cdots d\theta_s \\ &= \int_{[0,1]^n} \rho_1 \cdots \rho_s |J(t_1, \dots, t_n)| dt_1 \cdots dt_n \end{aligned}$$

$$\text{where } J = \frac{\pi^s x_1 \cdots x_r \rho_1 \cdots \rho_s}{t_{r+s}} \det(M) \text{ and } M = \begin{pmatrix} v_1^{(1)} & \cdot & \cdot & v_k^{(r+s)} \\ \cdot & \cdot & \cdot & \cdot \\ v_{r+s-1}^{(1)} & \cdot & \cdot & v_{r+s-1}^{(r+s)} \\ 1 & \cdot & \cdot & 2 \end{pmatrix} \text{ After calculating}$$

we get $\text{vol}(D_1^+) = \frac{1}{n} \pi^s |\det(M)|$.

We define $\text{Reg}(R) = \frac{1}{n} \det(M)$. So we have proved

$$\kappa = \frac{2^{r+s} \pi^s \text{Reg}(R)}{|W| \sqrt{|\text{disc}(R)|}}$$

For imaginary quadratic case there is no free part,so by default

$$Reg(R) = 1 \text{ and } s = 1, r = 0$$

.So

$$\begin{aligned} \kappa &= \frac{\pi}{\sqrt{|disc(R)|}} \text{ for } d \neq (-1), (-3) \\ &= \frac{\pi}{2\sqrt{|disc(R)|}}, \frac{\pi}{3\sqrt{|disc(R)|}} \text{ for } d = (-1), (-3) \end{aligned}$$

For real quadratic field there is a free part of rank 1 say u.And $r = 2, s = 0$.Now

$$u \mapsto (\sigma_1(u), \sigma_2(u)) \mapsto (\log|\sigma_1(u)|, \log|\sigma_2(u)|)$$

And

$$|\sigma_1(u)\sigma_2(u)| =$$

with

$$\sigma_1(u) = u \implies \log u(1, -1)$$

forms a lattice in Γ_U .So calculating we get

$$Reg(R) = \log(u)$$

and

$$\kappa = \frac{2\log(u)}{\sqrt{|disc(R)|}}$$

Chapter 3

Class Number Formula

3.1 Dedekind Zeta Function

Consider Dirichlet series $\sum_{n=1}^{\infty} \frac{a_n}{n^s}$ where $a_n \in \mathbb{C}$ and $s = x + iy$.

(3.1) Lemma Assume $\sum_{n \leq t} a_n$ is $O(t^r)$ for some real $r \geq 0$. Then the Dirichlet's series converges for all $s = x + iy$ with $x > r$, and is an analytic (= holomorphic) function of s on that half-plane.

Proof. It suffices to show that the series converges uniformly on every compact subset of the half-plane[4]. For each s we calculate the sum $\sum_{n=m}^M a_n n^{-s}$. Setting $A_k = \sum_{n=1}^k a_n$ we have

$$\sum_{n=m}^M \frac{a_n}{n^s} = \sum_{n=m}^M \frac{A_n}{n^s} - \sum_{n=m}^M \frac{A_{n-1}}{n^s} = \frac{A_M}{M^s} - \frac{A_{m-1}}{m^s} + \sum_{n=m}^{M-1} A_n \left(\frac{1}{n^s} - \frac{1}{(n+1)^s} \right)$$

From the $O(t^r)$ condition $\exists B$ such that $|A_n| \leq Bn^r \forall n$. Thus,

$$\left| \sum_{n=m}^M \frac{a_n}{n^s} \right| \leq B \left(\frac{M^r}{|M^s|} + \frac{(m-1)^r}{|m^s|} + \sum_{n=m}^{M-1} n^r \left| \frac{1}{n^s} - \frac{1}{(n+1)^s} \right| \right)$$

We write $s = x + iy$ and observe that

$$\frac{1}{n^s} - \frac{1}{(n+1)^s} = s \int_n^{n+1} \frac{dt}{t^{s+1}},$$

we get

$$\left| \frac{1}{n^s} - \frac{1}{(n+1)^s} \right| \leq |s| \int_n^{n+1} \frac{dt}{|t^{s+1}|} = |s| \int_n^{n+1} \frac{dt}{t^{x+1}} \leq \frac{|s|}{n^{x+1}}$$

leading to

$$\left| \sum_{n=m}^M \frac{a_n}{n^s} \right| \leq B \left(M^{r-x} + m^{r-x} + |s| \sum_{n=m}^{M-1} n^{r-x-1} \right)$$

If m and $M \rightarrow \infty$, then we obtain that this expression $\rightarrow 0$ for any fixed $s = x + iy$ in the half-plane $x > r$; since the last sum on the right hand side is bounded by

$$\int_{m-1}^{\infty} t^{r-x-1} dt = \frac{(m-1)^{r-x}}{x-r}$$

for any $m > 1$. This implies convergence of the series $\sum_{n=1}^{\infty} a_n n^{-s}$ for $x > r$. Moreover this is uniform convergence on compact subsets of the half-plane $x > r$ as each that type of set is bounded (so that $|s| \leq B'$ for some B'). It is also bounded away from the line $x = r$ (so that $x - r \geq \epsilon$ for some $\epsilon > 0$). Hence we get

$$\left| \sum_{n=m}^{\infty} \frac{a_n}{n^s} \right| \leq B \left(m^{-\epsilon} + B' \frac{(m-1)^{-\epsilon}}{\epsilon} \right)$$

□

Applying the above proposition to Reimann zeta function $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$, it converges and analytic on half plane $x > 1$. Now we define Dedekind zeta function ζ_K of a number field K for $x > 1$ by

$$\zeta_K(s) = \sum_{n=1}^{\infty} \frac{j_n}{n^s}$$

where $j_n = \text{no. of } I \text{ of } R = \mathbb{A} \cap K \text{ with } ||I|| = n$.

We know $\sum_{n \leq t} j_n = i(t) = O(t)$ hence the (3.1) says ζ_K converges and is analytic on half plane $x > 1$. We know $\zeta(s)$ can be extended to a analytic function for $x > 0$ except at $s = 1$ where there is simple pole.

For extending $\zeta_K(s)$ we have

$$\zeta_K(s) = \sum_{n=1}^{\infty} \frac{j_n - h\kappa}{n^s} + h\kappa \zeta(s)$$

for $x > 1$, where h is ideal class number in R . Applying the $Prop^n$

$$\sum_{n \leq t} j_n - h\kappa = i(t) - h\kappa t = O(t^{1 - \frac{1}{[K:\mathbb{Q}]}})$$

Hence ζ_K is analytic for $x > 1 - \frac{1}{[K:\mathbb{Q}]}$ except for $s=1$ where it has simple pole. As ζ_K absolutely converges for $x > 1$ we can rearrange terms to get

$$\zeta_K(s) = \sum_I \frac{1}{||I||^s} \quad \text{for } x > 1$$

(3.2) Lemma .Let $a_1, a_2, \dots, a_n \in \mathbb{C}$ be a sequence s.t $1 + a_n \neq 0$ then $\prod_{n=1}^{\infty} (1 + a_n)$ converges absolutely iff $\sum_{n=1}^{\infty} |a_n|$ converges absolutely.

(3.3) Theorem Let $a_1, \dots, a_n \in \mathbb{C}$, $|a_i| < 1 \forall i$ and $\sum_{i=1}^{\infty} |a_i| < \infty$. Then

$$\prod_{i=1}^{\infty} (1 - a_i)^{-1} = 1 + \sum_{j=1}^{\infty} \sum_{(r_1, \dots, r_j)} a_1^{r_1} \dots a_j^{r_j}$$

Both the sum and product are absolutely convergent, hence are independent of the order of a_i .

Proofs. are in¹

Now every I has a unique factorization in R in terms of P prime ideals. Hence

$$\sum ||P||^{-s} = \sum ||P||^{-x} < \sum ||I||^{-x} < \infty$$

As $\zeta_K(x)$ converges for $x > 1$. Hence applying the theorem we get

$$\zeta_K(s) = \prod_P (1 - \frac{1}{||P||^s})^{-1} \quad \text{for } x > 1$$

Now if we look at $\frac{\zeta_K(s)}{\zeta(s)}$ then its first part is analytic at $s = 1$ so

$$\rho = \lim_{s \rightarrow 1} \frac{\zeta_K(s)}{\zeta(s)} = h\kappa.$$

We will calculate h for abelian extension K , so K is galois extension. For each prime

$p \in \mathbb{Z}$, we denote r_p as the number of primes in R lying over p , and let the inertial degree $f(P|p)$ be denoted as f_p . Then,

$$\zeta_K(s) = \prod_p \left(1 - \frac{1}{p^{f_p s}}\right)^{-r_p}, \text{ for } x > 1$$

3.2 Class Number Formula

We recall the definition of L -functions:

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}$$

where χ is the character modulo m , where $K \subset \mathbb{Q}(e^{\frac{2\pi i}{m}})$ (by Kronecker Weber Theorem[1]) which converges for $x > 0$ for non trivial character and for trivial character it converges for $x > 1$. So,

$$L(s, 1) = \zeta(s) \prod_{p|m} \left(1 - \frac{1}{p^s}\right)$$

. Let G be the Galois group of $K / \mathbb{Q}$. Now, $G \subset \mathbb{Z}_m^* \implies \hat{G} \subset \hat{\mathbb{Z}}_m^*$. Fixing prime $p \nmid m$, we also have formula

$$\prod_{\chi \in \hat{G}} \left(1 - \frac{\chi(p)}{p^s}\right) = \left(1 - \frac{1}{p^{f_p s}}\right)^{r_p}, \text{ for } x > 1$$

, from which it follows that

$$\frac{\zeta_K(s)}{\zeta(s)} = \prod_{p|m} \left(1 - \frac{1}{p^s}\right) \left(1 - \frac{1}{p^{f_p s}}\right)^{-r_p} \prod_{1 \neq \chi \in \hat{G}} L(s, \chi)$$

As, for $\chi \neq 1$, $L(s, \chi)$ is analytic for $\sigma > 0$, we have the required formula for ρ just by putting $s = 1$

$$\rho = \prod_{p|m} \left(1 - \frac{1}{p^s}\right) \left(1 - \frac{1}{p^{f_p s}}\right)^{-r_p} \prod_{1 \neq \chi \in \hat{G}} L(1, \chi)$$

(3.4)Theorem :Let χ be a non-trivial character modulo m . Then

$$L(1, \chi) = -\frac{1}{m} \sum_{k=1}^{m-1} \tau_k(\chi) \log(1 - \omega^{-k})$$

where $\tau_k(\chi) = \sum_{a \in \mathbb{Z}_m^*} \chi(a) \omega^{ak}$ and $\log(1 - z) = -\sum_{n=1}^{\infty} z^n/n$ for all $z < 1$ and also for those z on unit circle for which series converges.

Proof For s in half plane $x > 1$ we have,

$$\begin{aligned} L(s, \chi) &= \sum_{a \in \mathbb{Z}_m^*} \chi(a) \sum_{n \equiv a \pmod{m}, n \geq 1} \frac{1}{n^s} \\ &= \sum_{a \in \mathbb{Z}_m^*} \chi(a) \sum_{n=1}^{\infty} \frac{\frac{1}{m} \sum_{k=0}^{m-1} \omega^{(a-n)k}}{n^s} \\ &= \frac{1}{m} \sum_{k=0}^{m-1} \tau_k(\chi) \sum_{n=1}^{\infty} \frac{\omega^{-nk}}{n^s} \end{aligned}$$

by absolute convergence exchange of sum possible. Also, $\tau_0(\chi) = \sum_{a \in \mathbb{Z}_m^*} \chi(a) = 0$, so $1 \leq k \leq m-1$. And for such k ,

$$\sum_{n=1}^{\infty} \frac{\omega^{-nk}}{n^s}$$

converges to analytic function on $x > 0$ and so cont. at $s = 1$ and equal to $-\log(1 - \omega^{-k})$. So we get the theorem just by approaching 1 from right. $\square$

We can further simplify the R.H.S. in the previous theorem using the values of τ_k , as given by the following lemma

(3.5) Lemma :Let χ be primitive character modulo m . Then

$$\tau_k(\chi) = \begin{cases} \bar{\chi}(k) \tau(k), & \text{if } (k, m) = 1 \\ 0, & \text{if } (k, m) > 1 \end{cases}$$

where $\tau(k) = \tau_1(k)$.

Proof. in [1]

Substituting the values of τ_k in the expression for $L(1, \chi)$, and calculating for odd and

even characters separately, we get

$$|L(1, \chi)| = \begin{cases} \frac{2}{\sqrt{m}} \left| \sum_{k \in \mathbb{Z}_m^*; k < m/2} \chi(k) \log \sin \frac{k\pi}{m} \right|, & \chi \text{ is even} \\ \frac{\pi}{|2 - \chi(2)|\sqrt{m}} \left| \sum_{k \in \mathbb{Z}_m^*} \chi(k) \right|, & \chi \text{ is odd} \end{cases}$$

This gives us the complete formula.

3.3 Quadratic Case

We will calculate the class number formula for the field $K = \mathbb{Q}[\sqrt{d}]$, where d is a square-free integer. We assume that $m = |\text{disc}(R)|$. So,

$$m = \begin{cases} |d|, & \text{if } d \equiv 1 \pmod{4} \\ |4d|, & \text{if } d \not\equiv 1 \pmod{4} \end{cases}$$

Also we can prove that $K \subset \mathbb{Q}(e^{2\pi i/m})$. Also, p is ramified in R if and only if $p \mid m$.

We note that in this case, $r_p = 1 = f_p$. Hence, using the formula of ρ ,

$$\rho = \prod_{p \mid m} \left(1 - \frac{1}{p}\right) \left(1 - \frac{1}{p^{f_p}}\right)^{-r_p} \prod_{1 \neq \chi \in \hat{G}} L(1, \chi) = L(1, \chi)$$

where χ is the unique non-trivial character modulo m corresponding to the character of the Galois group of K over $\mathbb{Q}$.

Let us define the character as following: for odd prime $p \nmid m$,

$$\chi(p) = \left(\frac{d}{p}\right)$$

where the R.H.S. is the Legendre Symbol. If $2 \nmid m$, which means that $d \equiv 1 \pmod{4}$,

$$\chi(2) = \begin{cases} 1, & \text{if } d \equiv 1 \pmod{8} \\ -1, & \text{if } d \equiv 5 \pmod{8} \end{cases}$$

The above two definitions taken together give us the multiplicative homomorphism $\chi : \mathbb{Z}_m^* \rightarrow \mathbb{C}$, defined by $\chi(n) = \left(\frac{d}{n}\right)$, for odd integers n , where $(.)$ denotes the Jacobi sym-

bol.

(3.6) Theorem : For Quadractic fields,we have if p is odd prime
If $p|d$,then

$$pR = (p, \sqrt{d})^2$$

If $p \nmid d$,then

$$pR = \begin{cases} (p, n + \sqrt{d})(p, n - \sqrt{d}) & \text{if } d \equiv n^2 \pmod{p} \\ \text{prime} & \text{if } d \not\equiv n^2 \pmod{p} \end{cases}$$

Proof.Given in⁴

Applying the above theorem for $p \nmid d$ i.e $p \nmid m$ (as d and m has same prime factors)
case,we get $r_p = 2, f_p = 1$ for $\left(\frac{d}{p}\right) = 1$ and $r_p = 1, f_p = 2$ for $\left(\frac{d}{p}\right) = -1$.Hence,

$$\begin{aligned} \left(1 - \frac{1}{p^{f_p s}}\right)^{-r_p} &= \left(1 - \frac{1}{p^s}\right)^{-1} \left(1 - \frac{\left(\frac{d}{p}\right)}{p^s}\right)^{-1} \\ \implies \zeta_K(s) &= \zeta(s) \prod_P \left(1 - \frac{\left(\frac{d}{p}\right)}{p^s}\right)^{-1} \\ \implies \frac{\zeta_K(s)}{\zeta(s)} &= L(s, \chi) \end{aligned}$$

Similar for $p = 2$ case we can verify the definition.Also χ here comes out to be primitive character mod m hence we can apply the previous thm in calculation.Moreover, χ is even if $d > 0$ and odd if $d < 0$.

So final class formula for quadratic case can be given by $h = \frac{\rho}{\kappa}$,

$$h = \begin{cases} \frac{1}{\log u} \left| \sum_{k \in \mathbb{Z}_m^*; k < m/2} \chi(k) \log \sin \frac{k\pi}{m} \right|, & d > 0 \\ \frac{1}{|2 - \chi(2)|\sqrt{m}} \left| \sum_{k \in \mathbb{Z}_m^*} \chi(k) \right|, & d < 0, d \neq -1, -3 \end{cases}$$

For $d = -1, -3$ case we get additional factor 2,3 resp. with the given formula.

Examples

1) $d = -2$: Here $m = 8$ as $d \not\equiv 1 \pmod{4}$. $\chi(3) = \left(\frac{-2}{3}\right) = \left(\frac{1}{3}\right)$. Hence,

$$h = \frac{1}{2}|\chi(1) + \chi(3)| = 1$$

2) $d = -5$: Here $m = 20$, $\chi(3) = \left(\frac{-5}{3}\right) = \left(\frac{1}{3}\right) = 1$, $\chi(7) = \left(\frac{-5}{7}\right) = \left(\frac{2}{7}\right) = 1$, $\chi(9) = \left(\frac{-5}{9}\right) = 1$

$$h = \frac{1}{2}|\chi(1) + \chi(3) + \chi(7) + \chi(9)| = 2$$

3) $d = 2$: Here $m = 8$, $\chi(3) = -1$, $u = (1 + \sqrt{2})$

$$h = \frac{1}{\log(1+\sqrt{2})}|\chi(1)\log\sin\left(\frac{\pi}{8}\right) + \chi(3)\log\sin\left(\frac{3\pi}{8}\right)| = 1$$

Since,

$$(1 + \sqrt{2}) = \frac{\sin\frac{3\pi}{8}}{\sin\frac{\pi}{8}}$$

4) $d = 3$: Here $m = 12$, $u = (2 + \sqrt{3})$, $\chi(5) = \left(\frac{3}{5}\right) = -1$

$$h = \frac{1}{\log(2+\sqrt{3})}|\chi(1)\log\sin\left(\frac{\pi}{12}\right) + \chi(5)\log\sin\left(\frac{5\pi}{12}\right)| = 1$$

Chapter 4

Localization

4.1 Definition and Construction

(4.1) Definition .Take A to be any commutative ring (unital).

We define the multiplicative subset S of A as a subset that is closed under multiplication and also $1 \in S$ and $0 \notin S$ (otherwise it will be trivial).

The **localization of A** with respect to S is a new ring AS^{-1} where elements are fractions with numerators in R and denominators in S . With the ring homomorphism $f : A \rightarrow AS^{-1}$ that maps S into $(AS^{-1})^\times$, it satisfies the universal property: if $g : A \rightarrow B$ is a ring homomorphism with $g(S) \subseteq B^\times$ then there is a unique ring homomorphism $h : AS^{-1} \rightarrow B$ s.t $g = h \circ f$

One also says that g factors uniquely through AS^{-1} (through f). This universal properties, guarantees that AS^{-1} is unique (upto isomorphism), provided that it exists.

(4.2) Construction .We construct AS^{-1} as the quotient of $A \times S$ modulo the equivalence relation

$$(a, s) \sim (b, t) \Leftrightarrow \exists u \in S \text{ such that } (at - bs)u = 0.$$

Denote the equivalence class of (a, s) as a/s . We find that AS^{-1} is a ring with additive identity $0/1$ and multiplicative identity $1/1$. Define $f(a) := a/1$; then $f : A \rightarrow AS^{-1}$ is a ring homomorphism.

For $s \in S$ we get $f(s)^{-1} = 1/s$, as $(s/1)(1/s) = s/s = 1/1 = 1$, thus $f(S) \subseteq (AS^{-1})^\times$.

since $\frac{q}{s} = a \in \mathfrak{p} \cap A \implies q = as \in \mathfrak{q}$, since $\mathfrak{q}$ is a prime ideal $a \in \mathfrak{q}$ because $s \notin \mathfrak{q}$.

For converse, let $\mathfrak{p}$ be any prime ideal of AS^{-1} . Then $\mathfrak{p} = \mathfrak{q} \cap A$ is a prime ideal of A . For checking $\mathfrak{p}$ is prime let $qq' \in \mathfrak{q} \implies \frac{qq'}{1} \in \mathfrak{p} \implies \frac{q}{1} \text{ or } \frac{q'}{1} \in \mathfrak{p} \implies$ either q or $q' \in \mathfrak{q}$. Also, $\mathfrak{q} \subseteq A \setminus S$. Because, if q were to contain an $s \in S$, then we will have $1 = s \cdot \frac{1}{s} \in \mathfrak{p}$ as $\frac{1}{s} \in AS^{-1}$. Moreover, one has

$$\mathfrak{p} = \mathfrak{q}S^{-1}.$$

For if $\frac{a}{s} \in \mathfrak{p}$, then $a = \frac{a}{s} \cdot s \in \mathfrak{p} \cap A = \mathfrak{q}$, hence $\frac{a}{s} = a \frac{1}{s} \in \mathfrak{q}S^{-1}$. The mappings $\mathfrak{q} \mapsto \mathfrak{q}S^{-1}$ and $\mathfrak{p} \mapsto \mathfrak{p} \cap A$ are therefore inverses of each other, and hence proved. $\square$

Examples : 1.) Let $\mathfrak{p}$ be a prime ideal of A . Then $S = A - \mathfrak{p}$ is multiplicatively closed (also $A - \mathfrak{p}$ is multiplicatively closed $\Leftrightarrow \mathfrak{p}$ is prime). We write $A_{\mathfrak{p}}$ instead of $S^{-1}A$ in this case. Here $\mathfrak{p}S^{-1}$ form an ideal $\mathfrak{m}$ in $A_{\mathfrak{p}}$. If $b/t \notin \mathfrak{m}$, then $b \notin \mathfrak{p}$, hence $b \in S$ and therefore b/t is a unit in $A_{\mathfrak{p}}$. Hence we get that if I is an ideal in $A_{\mathfrak{p}}$ and $I \not\subseteq \mathfrak{m}$, then I contains a unit and so it's the whole ring. Thus $\mathfrak{m}$ is the unique maximal ideal in $A_{\mathfrak{p}}$; hence, $A_{\mathfrak{p}}$ is a local ring. The process of this passing from A to $A_{\mathfrak{p}}$ is called localization at $\mathfrak{p}$.

2.) Take S as the complement of a union $\bigcup_{\mathfrak{p} \in X} \mathfrak{p}$ over a set X of prime ideals of A . In this case we write

$$A(X) = \left\{ \frac{f}{g} \mid f, g \in A, g \not\equiv 0 \pmod{\mathfrak{p} \text{ for } \mathfrak{p} \in X} \right\}$$

instead of AS^{-1} . The prime ideals of $A(X)$ correspond by (4.4) 1-1 to the prime ideals of A which are contained in $\bigcup_{\mathfrak{p} \in X} \mathfrak{p}$, all the others are being eliminated when passing from A to $A(X)$. When A is Dedekind Domain, if X is finite, then only the prime ideals from X survive in $A(X)$.

3.) Let $f \in A$ and let $S = \{f^n\}_{n \geq 0}$. We write A_f for AS^{-1} in this case

4.) If $A = \mathbb{Z}$, $\mathfrak{p} = (p)$, p a prime number; then,

$$A_{\mathfrak{p}} = \mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, p \nmid b \right\}$$

(4.5) Proposition. Consider $A_{\mathfrak{p}}$ with the unique maximal ideal, $\mathfrak{m}_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$. Then $\exists$ a natural

embedding

$$A/\mathfrak{p} \longrightarrow A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}},$$

identifying $A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}$ with the field of fractions of $A/\mathfrak{p}$. In particular, if $\mathfrak{p}$ is a maximal ideal of A , then we have

$$A/\mathfrak{p}^n \cong A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}^n \quad \text{for } n \geq 1.$$

Proof: Let us consider the homomorphism

$$f : A/\mathfrak{p}^n \longrightarrow A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}^n, \quad a \bmod \mathfrak{p}^n \longmapsto a \bmod \mathfrak{m}_{\mathfrak{p}}^n.$$

For $n = 1$, f is injective because $\mathfrak{p} = \mathfrak{m}_{\mathfrak{p}} \cap A$. Now $A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}$ is a field, hence it becomes the field of fractions of $A/\mathfrak{p}$. Let $\mathfrak{p}$ be maximal and $n \geq 1$. For every $s \in A \setminus \mathfrak{p}$ we have the identity $\mathfrak{p}^n + sA = A$, i.e., $\bar{s} = s \bmod \mathfrak{p}^n$ is a unit in $A/\mathfrak{p}^n$. For $n = 1$ this comes from the maximality of $\mathfrak{p}$, and for $n \geq 1$ it follows by induction: $A = \mathfrak{p}^{n-1} + sA \Rightarrow \mathfrak{p} = \mathfrak{p}A = \mathfrak{p}(\mathfrak{p}^{n-1} + sA) \subsetneq \mathfrak{p}^n + sA \Rightarrow \mathfrak{p}^n + sA = A$.

For injectivity of f : let $a \in A$ be s.t $a \in \mathfrak{m}_{\mathfrak{p}}^n$, i.e., $a = b/s$ with $b \in \mathfrak{p}^n$, $s \notin \mathfrak{p}$. Then $as = b \in \mathfrak{p}^n$, so that $\bar{a}\bar{s} = 0$ in $A/\mathfrak{p}^n$, and hence $\bar{a} = 0$ in $A/\mathfrak{p}^n$.

For surjectivity of f : let $a/s \in A_{\mathfrak{p}}$, $a \in A$, $s \notin \mathfrak{p}$. Then by the above formula, $\exists$ an $a' \in A$ s.t $a \equiv a's \bmod \mathfrak{p}^n$. Therefore $a/s \equiv a' \bmod \mathfrak{p}^n A_{\mathfrak{p}}$, i.e., $a/s \bmod \mathfrak{m}_{\mathfrak{p}}^n$ lies in the image of f . $\square$

In a local ring with maximal ideal $\mathfrak{m}$, every element $a \notin \mathfrak{m}$ is a unit. Indeed, since the principal ideal (a) is not contained in any other maximal ideal, it has to be the whole ring. So we have

$$A^* = A \setminus \mathfrak{m}.$$

4.3 Discrete Valuation Rings

(4.6) Definition. A discrete valuation ring is a principal ideal domain $\mathcal{O}$ with a unique maximal ideal $\mathfrak{p} \neq 0$. (localring + PID)

The maximal ideal is of the form $\mathfrak{p} = (\pi) = \pi\mathcal{O}$, for some prime element π . Since every element not contained in $\mathfrak{p}$ is a unit, it lead us that that, up to associated elements, π is the only prime element of $\mathcal{O}$. Hence, every nonzero element of $\mathcal{O}$ can be written as $\varepsilon\pi^n$, for some $\varepsilon \in \mathcal{O}^*$, and $n \geq 0$. More generally, every element $a \neq 0$ of the field of fractions K can be uniquely written as

$$a = \varepsilon\pi^n, \quad \varepsilon \in \mathcal{O}^*, \quad n \in \mathbb{Z}.$$

The exponent n is called the valuation of a . It is denoted $v(a)$, and it is characterized by the equation

$$(a) = \mathfrak{p}^{v(a)}.$$

The valuation is a function

$$v : K^* \longrightarrow \mathbb{Z}$$

Extend it to K by the convention $v(0) = \infty$. It also satisfies the conditions

$$v(ab) = v(a) + v(b), \quad v(a+b) \geq \min\{v(a), v(b)\}.$$

The discrete valuation rings arise as localizations of Dedekind domains. This we get from following proposition.

(4.7) Proposition. If A is a Dedekind domain, and $S \subseteq A \setminus \{0\}$ is a multiplicative subset, then AS^{-1} is also a Dedekind domain.

Proof: Let I be an ideal of AS^{-1} and $\mathfrak{a} = I \cap A$. Then $I = \mathfrak{a}S^{-1}$, because if $\frac{a}{s} \in I$, $a \in A$ and $s \in S$, then one has $a = s \cdot \frac{a}{s} \in I \cap A = \mathfrak{a}$, so that $\frac{a}{s} = a \cdot \frac{1}{s} \in \mathfrak{a}S^{-1}$. As $\mathfrak{a}$ is finitely generated, so is I , i.e., AS^{-1} is noetherian. It follows from (4.4) that every prime ideal of AS^{-1} is maximal, because this holds in A (Take U_1, U_2 prime ideals of AS^{-1} s.t $U_1 \subset U_2 \subset AS^{-1}$. Hence their correspondent P_1, P_2 in $A \setminus S$ exists s.t $P_1 \subset P_2 \subset A \leftarrow \rightarrow$). Lastly, AS^{-1} is integrally closed, for if $x \in K$ ($f.f(A) = f.f(AS^{-1})$) satisfies the equation

$$x^n + \frac{a_1}{s_1}x^{n-1} + \cdots + \frac{a_n}{s_n} = 0$$

with coefficients $\frac{a_i}{s_i} \in AS^{-1}$, then multiplying it with the n -th power of $s = s_1 \cdots s_n$ shows that sx is integral over A , hence $sx \in A$ and therefore $x \in AS^{-1}$. This proves that AS^{-1} is a Dedekind domain.

□

(4.8) Proposition. Let A be a noetherian integral domain. Then A is a Dedekind domain

iff, for all prime ideals $\mathfrak{p} \neq 0$, the localizations $A_{\mathfrak{p}}$ are discrete valuation rings.

Proof: If A is a Dedekind domain, then $A_{\mathfrak{p}}$ is also Dedekind domain from last theorem. The maximal ideal $\mathfrak{m} = \mathfrak{p}A_{\mathfrak{p}}$ is the only nonzero prime ideal of $A_{\mathfrak{p}}$. Since, for any prime ideal $\mathfrak{q} \in A_{\mathfrak{p}}, \mathfrak{q} \subseteq \mathfrak{m}$ and $\mathfrak{q}$ maximal ($A_{\mathfrak{p}}$ is D.D) $\implies \mathfrak{p} = \mathfrak{q}$. Thus, choosing any $\pi \in \mathfrak{m} \setminus \mathfrak{m}^2$, one finds $(\pi) = \mathfrak{m}$. This is so since $\mathfrak{m}^2 \subset (\pi) \subseteq \mathfrak{m}$ and $A_{\mathfrak{p}}$ has unique factorization over ideals with only prime ideal $\mathfrak{m}$. Moreover, $\mathfrak{m}^n = (\pi^n)$. Thus $A_{\mathfrak{p}}$ is a principal ideal domain, and hence a discrete valuation ring.

Claim:

$$A = \bigcap_{\mathfrak{p}} A_{\mathfrak{p}} \quad \text{over non zero } \mathfrak{p}$$

We can easily see that $A \subseteq \bigcap_{\mathfrak{p}} A_{\mathfrak{p}}$ since $A \subseteq A_{\mathfrak{p}}$. For converse take $\frac{a}{b} \in \bigcap_{\mathfrak{p}} A_{\mathfrak{p}}$, with $a, b \in A$. Then,

$$I = \{x \in A \mid xa \in bA\}$$

is an ideal which can't be contained in any prime ideal of A . Since, for any $\mathfrak{p}$, we may write $\frac{a}{b} = \frac{c}{s}$ with $c \in A, s \notin \mathfrak{p}$, which implies $sa = bc$, hence $s \in I \setminus \mathfrak{p}$. As I is not contained in any maximal ideal, it implies that $I = A$, hence for $x = 1 \in I$; $a = 1 \cdot a \in bA \implies \frac{a}{b} \in A$. Hence we are done with claim.

Now, suppose that the $A_{\mathfrak{p}}$ are discrete valuation rings. Since they are principal ideal domains, they are integrally closed. So $A = \bigcap_{\mathfrak{p}} A_{\mathfrak{p}}$ is also integrally closed. Lastly from (4.4), it follows that every prime ideal $\mathfrak{p} \neq 0$ of A is maximal because this is so in $A_{\mathfrak{p}}$. Therefore A is a Dedekind domain. $\square$

Chapter 5

Inverse Limits

Inverse limits is a general construction that is applied in any category with products, but we will only use the inverse limits in familiar concrete categories or systems such as groups, rings, and topological spaces.

Definition . A **directed set** is a set I with a relation " $\leq$ " that is reflexive ($i \leq i$), anti-symmetric ($i \leq j \leq i \Rightarrow i = j$), and transitive ($i \leq j \leq k \Rightarrow i \leq k$), in which for every i, j , there exists a k such that $i \leq k$ and $j \leq k$.

Definition . An **inverse system** (projective system) in a category is a family of rings $\{X_i : i \in I\}$ indexed by a directed set I with a family of homomorphisms $\{f_{ij} : X_j \rightarrow X_i : i \leq j\}$ such that

- i) each f_{ii} is the identity,
- ii) $f_{ik} = f_{ij} \circ f_{jk}$ for all $i \leq j \leq k$.

Definition . Let (X_i, f_{ij}) be an inverse system in the system of rings. The **inverse limit** (or projective limit) of (X_i, f_{ij}) inverse system is defined as :

$$X := \varprojlim X_i := \left\{ x \in \prod_{i \in I} X_i : x_i = f_{ij}(x_j) \text{ for all } i \leq j \right\} \subseteq \prod_{i \in I} X_i$$

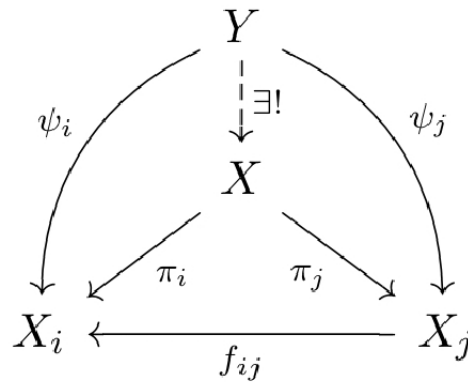
This inverse limit is a ring. The addition and multiplication is induced from the componentwise addition and multiplication in the product $\prod_{i \in I} X_i$.

Also, we get a family of homomorphism

$$\pi_i : X \rightarrow X_i$$

which we get by the restrictions of the projections $\prod X_i \rightarrow X_i$ satisfying $\pi_i = f_{ij} \circ \pi_j$ for $i \leq j$.

The inverse limit $X = \lim X_i$ has the universal property that if Y is another ring with the family of homomorphisms $\psi_i : Y \rightarrow X_i$ that satisfy $\psi_i = f_{ij} \circ \psi_j$ for $i \leq j$, then there is a unique homomorphism $h : Y \rightarrow X$ satisfying $\psi_i = \pi_i \circ h$.



Chapter 6

p-adic Numbers

6.1 Definitions

(6.1) Definition. Let p be a prime number. A **p-adic integer** can be defined as a formal infinite series as given below

$$a_0 + a_1p + a_2p^2 + \dots$$

where $0 \leq a_i < p, \forall i \geq 0$. This set is denoted as $\mathbb{Z}_p$

(6.2) p-adic Expansion. Let $r \in \mathbb{Z}_{(p)} (\subset \mathbb{Q})$, i.e $r = p^k \frac{n}{d}$ where $k \geq 0$ and $(nd, p) = 1$. We expand r similarly as we do in case of decimal expansion. Here we denote the integer k as the p-adic valuation of r ($v_p(r)$) and p^{-k} as its p-adic absolute value ($|r|_p$). The division step comprises of writing

$$r = ap^k + r'$$

where a is an integer s.t $0 \leq a < p$ and $r' = 0$ or a rational number s.t $|r'| < p^{-k}$ (i.e $v_p(r') > k$). We compute this terms using the bezout identity where since d and p are coprime, $\exists$ integers t and u s.t $td + up = 1$. So

$$r = p^k \frac{n}{d} (td + up) = p^k nt + p^{k+1} \frac{un}{d}$$

Then breaking down nt by p using euclidean division gives

$$nt = qp + a$$

with $0 \leq a < p$. Hence

$$r = ap^k + p^{k+1} \frac{qd + un}{d}$$

with r' as the second term which is rational with $|r'|_p \leq p^{-k}$. So, the p-adic expansion of r is the formal infinite series

$$r = \sum_{i=k}^{\infty} a_i p^i$$

which we obtained by repeating this process indefinitely. The division step is unique because

$$p^k a_1 + p^{k+1} r_1 = p^k a_2 + p^{k+1} r_2 \implies a_1 - a_2 = p(r_1 - r_2) \implies p | (a_1 - a_2)$$

Since $|a_1 - a_2| < p \implies a_1 = a_2$. Hence, p-adic expansion of r is unique. Now we extend this expansion to any rational number ' r ' given by $r = p^{-m} \frac{g}{h}$ where $g, h \in \mathbb{Z}$, p does not divide g, h . So, if

$$a_0 + a_1 p + a_2 p^2 + \dots$$

is the p-adic expansion of $\frac{g}{h} \in \mathbb{Z}_{(p)}$, then we say r has the p-adic expansion

$$a_0 p^{-m} + a_1 p^{-m+1} + \dots + a_m + a_{m+1} p + \dots$$

which we say as the p-adic number which we are going to define.

(6.3) Definition A p-adic Number is the formal infinite series

$$a_{-m} p^{-m} + a_{-m+1} p^{-m+1} + \dots + a_0 + a_1 p + \dots$$

where $m \in \mathbb{Z}$ and $0 \leq a_i < p, \forall i \geq -m$. This set is denoted as $\mathbb{Q}_{(p)}$

Now here we get a natural mapping from $\mathbb{Q} \hookrightarrow \mathbb{Q}_{(p)}$ which is injective as the expansion

is unique. So it also takes $\mathbb{Z} \hookrightarrow \mathbb{Z}_{(p)}$. Hence for every rational number r we get an identity,

$$r = \sum_{i=-m}^{\infty} a_i p^i$$

6.2 Algebraic Approach

Now we will view the p -adic integers with another representation. Let $f = \sum_{i=0}^{\infty} a_i p^i$ be a p -adic integer. Let

$$s_n = \sum_{i=0}^{n-1} a_i p^i \in \mathbb{Z}$$

. Now define the sequences of the residue classes as

$$\bar{s}_n \equiv s_n \pmod{p^n} \in \mathbb{Z}/p^n \mathbb{Z}$$

We also observe that

$$\bar{s}_{n+1} \equiv \bar{s}_n \pmod{p^n} \in \mathbb{Z}/p^n \mathbb{Z}$$

This are natural surjective homomorphism given by

$$\mathbb{Z}/p\mathbb{Z} \xleftarrow{\lambda_1} \mathbb{Z}/p^2\mathbb{Z} \xleftarrow{\lambda_2} \mathbb{Z}/p^3\mathbb{Z} \xleftarrow{\lambda_3} \dots$$

where

$$\lambda_n(\bar{s}_{n+1}) = \bar{s}_n$$

This forms the **projective limit** of the rings $\mathbb{Z}/p^n \mathbb{Z}$, denoted by $\varprojlim \mathbb{Z}/p^n \mathbb{Z}$ ($= \tilde{\mathbb{Z}}_p$ say). So we get,

$$\tilde{\mathbb{Z}}_p = \varprojlim \mathbb{Z}/p^n \mathbb{Z} = \{(x_n)_{n \in \mathbb{N}} \in \prod_{n=1}^{\infty} \mathbb{Z}/p^n \mathbb{Z} \mid \lambda_n(x_{n+1}) = x_n, n = 1, 2, 3, \dots\}$$

(6.4) Theorem $\mathbb{Z}_p \cong \tilde{\mathbb{Z}}_p$. (bijection)

Proof. Consider the function

$$\begin{aligned}\Phi : \mathbb{Z}_p &\longrightarrow \widetilde{\mathbb{Z}}_p, \\ a &\longmapsto (\phi_1(a), \phi_2(a) \dots)\end{aligned}$$

where for $i \geq 1$ and $a = \sum_{k=0}^{\infty} a_k p^k$ (it's p -adic expansion), we have

$$\begin{aligned}\phi_i : \mathbb{Z}_p &\longrightarrow \mathbb{Z}/p^i\mathbb{Z} \\ a &\longmapsto \sum_{k=0}^{i-1} a_k p^k \pmod{p^i}.\end{aligned}$$

Now, choose $(a_1, a_2, \dots) \in \widetilde{\mathbb{Z}}_p$, and for $n \geq 1$, let $0 \leq s_n \leq p^n - 1$ be the unique integer such that $s_n \equiv a_n \pmod{p^n}$. In base p , write

$$s_n = \lambda_{n,0} + \lambda_{n,1}p + \dots + \lambda_{n,n-1}p^{n-1}.$$

For $0 \leq i \leq n-1$, we have $\lambda_{n+1,i} = \lambda_{n,i} \Rightarrow \lambda_{n,i} \rightarrow \lambda_i$ (constant sequence) for some λ_i , as $n \rightarrow \infty$. Using this, let

$$\begin{aligned}\Psi : \widetilde{\mathbb{Z}}_p &\longrightarrow \mathbb{Z}_p \\ (a_1, a_2, \dots) &\longmapsto \sum_{k=0}^{\infty} \lambda_k p^k.\end{aligned}$$

We also get that $\Phi \circ \Psi$ and $\Psi \circ \Phi$ provides identity maps on $\widetilde{\mathbb{Z}}_p$ and $\mathbb{Z}_p$, hence proving the theorem. $\square$

Thus, we get a ring structure of $\mathbb{Z}_p$ that it inherits from $\widetilde{\mathbb{Z}}_p$

6.3 Some properties

(6.5) Lemma.: $\frac{1}{1+kp} = 1 - kp + kp^2 + \dots$ where $k \in \mathbb{Z}_p$

Proof. Let $s_n = 1 - kp + (kp)^2 - (kp)^3 + \dots + (kp)^{n-1} \in \mathbb{Z}_p$. Then we have the identity,

$$\begin{aligned} 1 &= s_n(1 + kp) + (kp)^n(-1)^n \\ \Rightarrow s_n(1 + kp) - 1 &= (kp)^n(-1)^{n+1} \\ \Rightarrow |s_n(1 + kp) - 1|_p &< \frac{1}{p^n} \leq \frac{1}{p^N} \quad \forall n \geq N \\ \Rightarrow \lim_{n \rightarrow \infty} s_n(1 + kp) &= 1 \quad (\text{wrt } |\cdot|_p) \\ \Rightarrow (1 + kp) \lim_{n \rightarrow \infty} s_n &= 1 \end{aligned}$$

Now $s = \lim_{n \rightarrow \infty} s_n = 1 - kp + (kp)^2 + \dots \in \mathbb{Z}_p$. Hence,

$$(1 + kp)^{-1} = 1 - kp + (kp)^2 - (kp)^3 + \dots$$

□

(6.6) Proposition. The unit group $\mathbb{Z}_p^\times$ in the ring $\mathbb{Z}_p$ consists of the p -adic integers

$$\mathbb{Z}_p^\times = \left\{ \sum_{i \geq 0} a_i p^i : a_0 \neq 0 \right\}$$

Proof. Take any $x \in \mathbb{Z}_p^\times$. Since $\mathbb{Z}_p \cong \tilde{\mathbb{Z}}_p$, we can say, $x = (s_0, s_1, s_2, \dots, s_n, \dots)$. Let, $y = x^{-1} = (t_0, t_1, t_2, \dots, t_n, \dots)$ as $y \in \mathbb{Z}_p^\times$. Then,

$$yx = (s_0 t_0, s_1 t_1, s_2 t_2, \dots) = (1, \dots, 1) \Rightarrow s_0 t_0 \equiv 1 \pmod{p}$$

Now $s_0 = a_0$. So, $a_0 t_0 \equiv 1 \pmod{p} \Rightarrow a_0 \neq 0$. This proves the inclusion $\mathbb{Z}_p^\times \subset \left\{ \sum_{i \geq 0} a_i p^i : a_0 \neq 0 \right\}$. Conversely, choose $0 < b_0 < p$ with $a_0 b_0 \equiv 1 \pmod{p}$ and write $a_0 b_0 = 1 + kp$. Hence, if we write $a = a_0 + p\alpha$, where α is also a p -adic integer, then

$$a \cdot b_0 = 1 + kp + p\alpha b_0 = 1 + p\kappa$$

for some p -adic integer κ . It is enough to show that the p -adic integer $1 + \kappa p$ is unit, since then we can write

$$a \cdot b_0(1 + \kappa p)^{-1} = 1 \implies a^{-1} = b_0(1 + \kappa p)^{-1}.$$

Now, notice that from last lemma, we can take

$$(1 + \kappa p)^{-1} = 1 - \kappa p + (\kappa p)^2 - \dots = 1 + c_1 p + c_2 p^2 + \dots$$

with integers $c_i \in \{0, 1, \dots, p-1\}$. Hence proved. $\square$

(6.7) Corollary. Any p -adic integer can be written in a unique way as $a = up^k \in \mathbb{Z}_p$ where $u \in \mathbb{Z}_p^\times$ and $k \geq 0$

(6.8) Proposition. $\mathbb{Q}_p$ is the field of fractions of $\mathbb{Z}_p$ Proof. Clearly for any $f \in \mathbb{Q}_p$ we can write ,

$$f = p^{-m}g \quad \text{where } g \in \mathbb{Z}_p$$

Hence, $\mathbb{Q}_p \subseteq f.f(\mathbb{Z}_p)$. Now,

$$f.f(\mathbb{Z}_p) = \left\{ \frac{r}{s} \mid r, s \in \mathbb{Z}_p, s \neq 0 \right\}$$

. By the last corollary we can say $s = up^k$ with u as unit and $k \geq 0$. In particular,

$$\frac{r}{s} = \frac{r}{up^k} = \frac{u^{-1}r}{u^{-1}up^k} = \frac{u^{-1}r}{p^k}$$

Now $u^{-1}r \in \mathbb{Z}_p$ and hence proved. $\square$

Chapter 7

Absolute Values of p-adic

7.1 Definition and Properties

(7.1) Definition. If $0 \neq x \in \mathbb{Z}$, we define the p -adic valuation (or exponential valuation) of x as

$$v_p(x) = \max \{r : p^r \mid x\} \geq 0.$$

For $a/b \in \mathbb{Q}$, the p -adic valuation of a/b

$$v_p : \mathbb{Q} \longrightarrow \mathbb{Z} \cup \{\infty\}$$

$$v_p\left(\frac{a}{b}\right) = v_p(a) - v_p(b).$$

Notice that for a rational a/b , the value of $v_p(a/b)$ is well defined, i.e., if $a/b = a'/b'$ then

$$v_p a - v_p b = v_p a' - v_p b'.$$

We also put the convention that $v_p(0) = \infty$.

(7.2) Proposition. If $x, y \in \mathbb{Q}$, then v_p satisfies:

- (i) $v_p(x) = \infty$ iff $x = 0$;
- (ii) $v_p(xy) = v_p(x) + v_p(y)$;
- (iii) $v_p(x + y) \geq \min \{v_p(x), v_p(y)\}$ with equality if $v_p(x) \neq v_p(y)$.

Proof. Clearly (i) and (ii) are true, we will only prove (iii). Let x, y be non-zero rational numbers. Write $x = p^r \frac{a}{b}$ and $y = p^s \frac{c}{d}$, where $a, b, c, d \in \mathbb{Z}$ with $p \nmid a, b, c, d$ and $r, s \in \mathbb{Z}$.

Now if $r = s$, we have

$$\begin{aligned} x + y &= p^r \left(\frac{a}{b} + \frac{c}{d} \right) \\ &= p^r \frac{(ad + bc)}{bd} \end{aligned}$$

which gives $v_p(x + y) \geq r$ since $p \nmid bd$. Now suppose that $r \neq s$, say $s > r$. Then

$$\begin{aligned} x + y &= p^r \left(\frac{a}{b} + p^{s-r} \frac{c}{d} \right) \\ &= p^r \frac{(ad + p^{s-r}bc)}{bd} \end{aligned}$$

Observe that as $s - r > 0$ and $p \nmid ad$, then

$$v_p(x + y) = r = \min \{v_p(x), v_p(y)\}$$

For the case when at least one of the terms is 0, we can clearly see that $v_p(x + y) = \min \{v_p(x), v_p(y)\}$

(7.3) Definition. For $x \in \mathbb{Q}$, define the **p -adic absolute value** (multiplicative valuation) of x as

$$|x|_p = \begin{cases} p^{-v_p(x)} & \text{if } x \neq 0, \\ p^{-\infty} = 0 & \text{if } x = 0 \end{cases}$$

(7.4) Proposition. The function $|\cdot|_p : \mathbb{Q} \longrightarrow \mathbb{R}^+$ has the properties

- (a) $|x|_p = 0$ if and only if $x = 0$;
- (b) $|xy|_p = |x|_p |y|_p$;
- (c) $|x + y|_p \leq \max \{|x|_p, |y|_p\}$, with equality if $|x|_p \neq |y|_p$.
- (d) $|x + y|_p \leq |x|_p + |y|_p$

Hence it satisfies the conditions of norm on $\mathbb{Q}$

Proof. This follows from Proposition 7.3.

We will show that the absolute values $|\cdot|_p$ and $|\cdot|$ are all the possible norms on $\mathbb{Q}$: any other norm is a power $|p|^s$ or $|\cdot|^s$, for some real number $s > 0$. The usual absolute value $|\cdot|$ is denoted here by $|\cdot|_\infty$. In addition to the absolute values $|\cdot|_p$, it satisfies the following property:

(7.5) Proposition. For every $a(\neq 0) \in \mathbb{Q}$, one has

$$\prod_p |a|_p = 1,$$

where p varies over all prime numbers as well as the symbol ∞ .

Proof: In the prime factorization

$$a = \pm \prod_{p \neq \infty} p^{v_p}$$

of a , the exponent v_p of p is the exponential valuation $v_p(a)$ and the sign equals $\frac{a}{|a|_\infty}$. So, we can rewrite the equation as

$$\begin{aligned} a &= \frac{a}{|a|_\infty} \prod_{p \neq \infty} \frac{1}{|a|_p}, \\ \implies \prod_p |a|_p &= 1 \end{aligned}$$

.

7.2 Analytic Approach

(7.6) Definition. A **Cauchy sequence** with respect to $|\cdot|_p$ in $\mathbb{Q}$ is defined by a sequence $\{x_n\}$ of rational numbers such that for every $\varepsilon > 0$, there exists a positive integer n_0 s.t

$$|x_n - x_m|_p < \varepsilon \text{ for all } n, m \geq n_0.$$

(7.7) Proposition. Every formal infinite series

$$\sum_{\nu=0}^{\infty} a_\nu p^\nu, \quad 0 \leq a_\nu < p,$$

gives us a Cauchy sequence via its partial sums

$$x_n = \sum_{\nu=0}^{n-1} a_\nu p^\nu,$$

Proof. For any $n, m \geq N$ we have,

$$|x_n - x_m|_p = \left| \sum_{i=m}^{n-1} a_i p^i \right|_p \leq \max_{m \leq i < n} \{ |a_i p^i|_p \} \leq \frac{1}{p^m} \leq \frac{1}{p^N}.$$

(7.8) Definition A sequence $\{x_n\}$ in $\mathbb{Q}$ is called a **nullsequence** with respect to $|\cdot|_p$ if x_n converges to 0 w.r.t $|\cdot|_p$ norm .(i.e $|x_n|_p$ sequence converging to 0 in the usual topology of $\mathbb{Q}$.)

(7.9) Proposition . Define the set of all Cauchy sequences of elements of $\mathbb{Q}$ as:

$$\mathcal{C} = \mathcal{C}_p(\mathbb{Q}) = \{(x_n) : (x_n) \text{ is a Cauchy sequence with respect to } |\cdot|_p\}$$

Define the ring operations as given

$$\begin{aligned} (x_n) + (y_n) &= (x_n + y_n) \\ (x_n) \cdot (y_n) &= (x_n y_n) \end{aligned}$$

Then $\mathcal{C}$ is a commutative ring with unity.

Proof. The additions and multiplications are commutative, associative and distributive, as it follows from $\mathbb{Q}$ which is a ring. The identity is the constant sequence $x_n = 1$. We just need to show that the sum and product of two Cauchy sequences is also a Cauchy sequence.

Here,

$$|(x_n + y_n) - (x_m + y_m)|_p \leq |x_n - y_n|_p + |x_m - y_m|_p \leq 2\epsilon \quad \forall m, n \geq N$$

Also a cauchy sequence is bounded, so we can get a bound of x_n and y_n by some large positive constants say A and B , respectively. Hence

$$\begin{aligned} |x_n y_n - x_m y_m| &= |x_n y_m - x_m y_n + x_m y_n - x_m y_m| \\ &= |y_n (x_n - x_m) + x_m (y_n - y_m)|. \end{aligned}$$

$$|x_n y_n - x_m y_m| \leq \left| B \left(\frac{\epsilon}{2B} \right) + A \left(\frac{\epsilon}{2A} \right) \right| = \epsilon \quad \forall n, m \geq M$$

completes the proof. □

We also observe that $\mathbb{Q} \subseteq \mathcal{C}$ by considering the map from some rational number x to a

corresponding constant sequence (x) in $\mathcal{C}$.

(7.10) Proposition. We define $\mathcal{N} \subset \mathcal{C}$ to be the set of nullsequences,

$$\mathcal{N} = \left\{ (x_n) : x_n \xrightarrow{|\cdot|_p} 0 \right\} = \left\{ (x_n) : \lim_{n \rightarrow \infty} |x_n|_p = 0 \right\}$$

and this set is a maximul ideal of the ring $\mathcal{C}$.

Proof. $\mathcal{N} \subset \mathcal{C}$ as any converging sequence is a cauchy sequence and it is an ideal as for any $(x_n), (y_n) \in \mathcal{N}$ and $(r_n) \in \mathcal{C}$ since $(x_n) + (y_n) = (x_n + y_n)$ and $(r_n) \cdot (x_n) = (r_n x_n)$ and also (r_n) is bounded, we get

$$x_n + y_n \xrightarrow{|\cdot|_p} 0$$

$$|r_n x_n|_p \leq B\epsilon \quad \text{for some bound } B > 0, \forall n \geq N$$

We have proved it is an ideal. Now we will prove its maximality. Let $e = (1, 1, 1, \dots)$.

Then e is the multiplicative identity of $\mathcal{C}$. Suppose $x \in \mathcal{C}, x \notin \mathcal{N}$.

To prove that $\mathcal{N}$ is a maximal ideal of $\mathcal{C}$, it is enough to show $(\mathcal{N}, x) = (e)$.

We know for any cauchy sequence with norm it is convergent iff it has a convergent subsequence. Now if x has a convergent null subsequence then x will be a null sequence. Since $x \notin \mathcal{N}$, at most finitely many terms of x are zero.

So, suppose $x_n \neq 0$ for $n > N$.

Let f be the element of $\mathcal{C}$ such that $f_k = 1$ for $0 \leq k \leq N$ and $f_k = 0$ for $k > N$. Clearly $f \in \mathcal{N}$.

Let $y = x - xf + f \implies y = f + x(1 - f) \implies y \in (\mathcal{N}, x)$

Also $y \notin \mathcal{N}$ as $x \not\xrightarrow{|\cdot|_p} 0$. By choice of f , all terms of y are nonzero which can be given by

$$y = (1, 1, 1, \dots, 1, x_n, x_{n+1}, \dots)$$

Let $1/y$ denote the sequence $(1/y_0, 1/y_1, 1/y_2, \dots)$.

Since y is a Cauchy sequence with all terms nonzero, and $y \notin \mathcal{N}$ implying no null subsequence of y , the terms of y are bounded away from zero (i.e $|y_n| \geq B \quad \forall n \geq N$), hence the products $|y_m y_n|_p \geq B^2 \quad \forall n \geq N$. Hence,

$$\left\| \frac{1}{y_m} - \frac{1}{y_n} \right\|_p = \left\| \frac{y_n - y_m}{y_m y_n} \right\|_p \leq \frac{\epsilon}{B^2} \quad \forall n, m \geq N$$

So we get that $1/y$ is a cauchy sequence that is $1/y \in \mathcal{C}$.

Now $y \in (\mathcal{N}, x)$ and $1/y \in \mathcal{C}$ that implies,

$$y(1/y) \in (\mathcal{N}, x) \implies (\mathcal{N}, x) = (e)$$

□

(7.11) Definition . Define the field of p -adic numbers to be the quotient of the ring $\mathcal{C}$ by its maximal ideal $\mathcal{N}$:

$$\mathbb{Q}_p = \mathcal{C}/\mathcal{N}$$

Here we have constructed a new ring $\mathbb{Q}_p$, based upon the p -adic norm $|\cdot|_p$ which we call as the **Completion** of $\mathbb{Q}$ (also denoted as $\hat{\mathbb{Q}}$).

(7.12)Definition. If λ is an element of $\mathbb{Q}_p$ and (x_n) is any Cauchy sequence in $\mathbb{Q}$ representing λ , we define

$$|\lambda|_p = \lim_{n \rightarrow \infty} |x_n|_p.$$

This limit exist as $\{|x_n|_p\}$ is a cauchy sequence in $\mathbb{R}$. Since from triangle inequality,

$$||x_n|_p - |x_m|_p| \leq |x_n - x_m|_p \leq \epsilon \quad \forall n \geq N$$

$|\lambda|_p$ is also well-defined. Consider two sequences (x_n) and (y_n) converging to the same limit. Then we have ,

$$\begin{aligned} (x_n) \sim (y_n) &\Rightarrow (x_n) - (y_n) \in \mathcal{N} \Rightarrow (x_n) - (y_n) = (z_n) \in \mathcal{N} \\ &\Rightarrow |x_n - y_n|_p = |z_n|_p \\ &\Rightarrow \lim_{n \rightarrow \infty} |x_n - y_n|_p = \lim_{n \rightarrow \infty} |z_n|_p = 0 \\ &\Rightarrow \lim_{n \rightarrow \infty} ||x_n|_p - |y_n|_p| = 0 \\ &\Rightarrow \lim_{n \rightarrow \infty} |x_n|_p = \lim_{n \rightarrow \infty} |y_n|_p \end{aligned}$$

Remark . Notice that for $a \in \mathbb{Q}$, the constant sequence $(a_n) = (a)$ is Cauchy sequence. So we obtain the element $\{a\}$ in $\mathbb{Q}_p$, that allows us to embed $\mathbb{Q}$ as a subring of $\mathbb{Q}_p$ (the inclusion $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$ preserves sums and products). We will identify $\mathbb{Q}$ with its image, hence we will use $a \in \mathbb{Q}$ to denote the element $\{a\} \in \mathbb{Q}_p$.

(7.13) Lemma . If x is an element of $\mathbb{Q}_p$ and (x_n) is any Cauchy sequence in $\mathbb{Q}$ repre-

sending x , then

$$x = \lim_{n \rightarrow \infty} x_n \quad \text{w.r.t } |\cdot|_p.$$

Proof. We know (x_n) is a Cauchy sequence in $\mathbb{Q}$ w.r.t $|\cdot|_p$. Hence,

$$\begin{aligned} |x_n - x_m|_p &< \varepsilon \quad \forall n, m \geq N \\ \Rightarrow \lim_{m \rightarrow \infty} |x_n - x_m|_p &< \varepsilon \quad \forall n \geq N \\ \Rightarrow |(x_n) - x|_p &< \varepsilon \quad \forall n \geq N \\ \Rightarrow \lim_{n \rightarrow \infty} x_n &= x \end{aligned}$$

Third step happens as $\{x - (x_n)\}_m = \{x_m - x_n\}_m$ for a fixed x_n □

(7.14) Theorem. $\mathbb{Q}_p$ is complete with respect to $|\cdot|_p$. Moreover, $\mathbb{Q}$ can be identified with a dense subring of $\mathbb{Q}_p$.

Proof. Take (α_n) as the Cauchy sequence in $\mathbb{Q}_p$ w.r.t the norm $|\cdot|_p$. Then we need to show that $\exists \alpha \in \mathbb{Q}$ for which

$$\lim_{n \rightarrow \infty} \alpha_n = \alpha \quad \text{w.r.t } |\cdot|_p$$

Note that each α_n is actually the equivalence class of some Cauchy sequence (a_{mn}) in $\mathbb{Q}$ w.r.t the norm $|\cdot|_p$, hence by last lemma,

$$\alpha_n = \lim_{m \rightarrow \infty} a_{mn} \quad \text{w.r.t } |\cdot|_p$$

We will construct a Cauchy sequence (c_n) in $\mathbb{Q}$ w.r.t $|\cdot|_p$ s.t

$$\{c_n\} = \lim_{n \rightarrow \infty} \alpha_n \quad \text{w.r.t } |\cdot|_p$$

That implies $\alpha = \{c_n\}$ is the required limit of (α_n) .

Now for each n , by last equation $\exists M_n$ s.t $\forall m \geq M_n$,

$$|\alpha_n - a_{mn}|_p < \frac{\varepsilon}{3}.$$

We define the sequence (c_n) by setting $c_n = a_n M_n$. We must show it has the required properties.

Claim 1: (c_n) is Cauchy w.r.t $|\cdot|_p$.

Proof. Let $\varepsilon > 0$. Since, (α_n) is Cauchy $\exists M'$ s.t $\forall i, j \geq M'$

$$|\alpha_i - \alpha_j|_p < \frac{\varepsilon}{3}.$$

Thus

$$\begin{aligned} |c_i - c_j|_p &= |(a_{iM_i} - \alpha_i) + (\alpha_i - \alpha_j) + (\alpha_j - a_{jM_j})|_p \\ &\leq |a_{iM_i} - \alpha_i|_p + |\alpha_i - \alpha_j|_p + |\alpha_j - a_{jM_j}|_p \end{aligned}$$

Choose $M = \max \{M', M_i, M_j\}$, then for $i, j \geq M$, we have

$$|c_i - c_j|_p < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon$$

and so the sequence (c_n) is Cauchy.

Claim 2: $\lim_{m \rightarrow \infty} \alpha_m = \{c_n\}$ w.r.t $|\cdot|_p$.

Proof. Let $\varepsilon > 0$. Denote $\{c_n\} = c$. We have

$$\begin{aligned} |(c - \alpha_m)|_p &= |((c - a_{mM_m}) + (a_{mM_m} - \alpha_m))|_p \\ &\leq |(c - a_{mM_m})|_p + |(a_{mM_m} - \alpha_m)|_p \\ &= \lim_{n \rightarrow \infty} |(a_{nM_n} - a_{mM_m})|_p + |(a_{mM_m} - \alpha_m)|_p \end{aligned}$$

Choose M'' s.t $M'' \geq 2/\varepsilon$ and s.t $\forall n, m \geq M''$ then

$$|(a_{nM_n} - a_{mM_m})|_p < \frac{\varepsilon}{2}.$$

So for $m, n > M''$ we have

$$|(a_{mM_m} - a_{nM_n})|_p + |(a_{mM_m} - \alpha_m)|_p < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

Hence we are done. □

(7.15) Lemma. Let $(x_n) \in \mathcal{C} \setminus \mathcal{N}$. The sequence of rational numbers $(|x_n|_p)$ is eventually constant, i.e, $\exists N$ such that $|x_n|_p = |x_m|_p$ whenever $m, n \geq N$.

Proof. Since (x_n) is a Cauchy sequence which does not tend to 0, that implies it has no convergent subsequence to 0 and only finite 0's are there in sequence (from proof of (7.10)). So, we can find c and N_1 such that

$$n \geq N_1 \implies |x_n|_p \geq c > 0.$$

Also $\exists N_2$ s.t

$$n, m \geq N_2 \implies |x_n - x_m|_p < c.$$

Let $N = \max \{N_1, N_2\}$. Then by Proposition (7.4)(c),

$$n, m \geq N_2 \implies |x_n - x_m|_p < \max \left\{ |x_n|_p, |x_m|_p \right\}$$

and so $|x_n|_p = |x_m|_p$. $\square$

Now, if $\{x_n\}$ is the representative of the Cauchy sequence for $x \in \mathbb{Q}_p$ where $x_n \neq 0$, then

$$v_p(x_n) = -\log_p |x_n|_p$$

either diverges to ∞ or is a Cauchy sequence in $\mathbb{Z}$ which eventually becomes constant for large n by (7.13) Lemma. Thus we can extend p -adic exponential valuation v_p on $\mathbb{Q}$ to an exponential valuation.

(7.16) Definition. Define **p -adic exponential** by

$$v_p : \mathbb{Q}_p \longrightarrow \mathbb{Z} \cup \{\infty\}$$

where

$$v_p(x) = \lim_{n \rightarrow \infty} v_p(x_n) = v_p(x_n) \quad \text{for } n \geq n_0$$

We also find that for all $x \in \mathbb{Q}_p$,

$$|x|_p = p^{-v_p(x)}.$$

(7.17) Proposition. Define the set

$$\mathbb{Z}_p := \left\{ x \in \mathbb{Q}_p \mid |x|_p \leq 1 \right\}$$

Then $\mathbb{Z}_p$ is a subring of $\mathbb{Q}_p$ with the group of units,

$$\mathbb{Z}_p^* = \left\{ x \in \mathbb{Z}_p \mid |x|_p = 1 \right\}.$$

Proof: $\mathbb{Z}_p$ is closed under addition and multiplication since for any $x, y \in \mathbb{Z}_p$

$$|x + y|_p \leq \max\{|x|_p, |y|_p\} \leq 1 \text{ and } |xy|_p = |x|_p |y|_p \leq 1.$$

For any unit $x \in \mathbb{Z}_p$, let y be the inverse. Now,

$$|xy|_p = 1 \implies |y|_p \geq \frac{1}{|x|_p} \geq 1$$

But $y \in \mathbb{Z}_p$ and so $|y|_p \leq 1$. Hence, $|x|_p = |y|_p = 1$. And so,

$$\mathbb{Z}_p^* = \left\{ x \in \mathbb{Z}_p \mid |x|_p = 1 \right\}.$$

(7.18) Corollary. Every element $x \in \mathbb{Q}_p^*$ (resp. $\mathbb{Z}_p$) admits a unique representation

$$x = p^m u \quad \text{with } m \in \mathbb{Z} \text{ (resp. } \in \{\mathbb{N} \cup (0)\}) \text{ and } u \in \mathbb{Z}_p^*.$$

Proof. If $v_p(x) = m \in \mathbb{Z}$, then $v_p(xp^{-m}) = 0$, hence $|xp^{-m}|_p = 1$, i.e., $u = xp^{-m} \in \mathbb{Z}_p^*$. For $x \in \mathbb{Z}_p, |x|_p \leq 1 \implies v_p(x) = m \geq 0$. Hence same calculations follows.

(7.19) Proposition. $\mathbb{Z}_p$ is the closure of the ring $\mathbb{Z}$ w.r.t $|\cdot|_p$ norm in the field $\mathbb{Q}_p$.

Proof. If $\{x_n\}$ is a convergent sequence (also Cauchy sequence) in $\mathbb{Z}$ then by (7.13), for some $x \in \mathbb{Q}_p$ $x = \lim_{n \rightarrow \infty} x_n$; then $|x_n|_p \leq 1 \implies |x|_p \leq 1$, hence $x \in \mathbb{Z}_p$. For Converse, let $x = \lim_{n \rightarrow \infty} x_n \in \mathbb{Z}_p$, for a Cauchy sequence $\{x_n\}$ in $\mathbb{Q}$. We observe that $|x|_p = |x_n|_p \leq 1$ for $n \geq n_0$ from (7.15). So, $x_n = \frac{a_n}{b_n}$, with $a_n, b_n \in \mathbb{Z}, (b_n, p) = 1 \forall n \geq n_0$. Choosing for each $n \geq n_0$ a solution $y_n \in \mathbb{Z}$ of the congruence $b_n y_n \equiv a_n \pmod{p^n}$ gives us $|x_n - y_n|_p \leq \frac{1}{p^n}$ and thus $x = \lim_{n \rightarrow \infty} y_n, \implies x$ belongs to the closure of $\mathbb{Z}$.

(7.20) Proposition. The nonzero ideals of the ring $\mathbb{Z}_p$ are the principal ideals

$$(p^n) = p^n \mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid v_p(x) \geq n\},$$

with $n \geq 0$. Hence with only one unique maximal ideal (p) , $\mathbb{Z}_p$ is a local ring and so Discrete valuation ring. Also one has

$$\mathbb{Z}_p / p^n \mathbb{Z}_p \cong \mathbb{Z} / p^n \mathbb{Z}$$

Proof: Let $I \neq (0)$ be an ideal of $\mathbb{Z}_p$ and $x = p^m u, u \in \mathbb{Z}_p^*$, an element of I with smallest possible m (since $|x|_p \leq 1$, one has $m \geq 0$). Then $I = p^m \mathbb{Z}_p$ as for any $y = p^n u' \in I, u' \in \mathbb{Z}_p^*$, implies $n \geq m$, hence $y = (p^{n-m} u') p^m \in p^m \mathbb{Z}_p$ and since $(x) = p^m \mathbb{Z}_p \subseteq I$. Now, the homomorphism

$$\mathbb{Z} \longrightarrow \mathbb{Z}_p / p^n \mathbb{Z}_p, \quad a \longmapsto a \bmod p^n \mathbb{Z}_p,$$

has kernel $p^n \mathbb{Z}$ and is surjective. As, for every $x \in \mathbb{Z}_p$, there exists by (7.19) an $a \in \mathbb{Z}$ s.t

$$|x - a|_p \leq \frac{1}{p^n}$$

i.e., $v_p(x - a) \geq n$, so $x - a \in p^n \mathbb{Z}_p$ and thus $x \equiv a \bmod p^n \mathbb{Z}_p$. So we get the isomorphism

$$\mathbb{Z}_p / p^n \mathbb{Z}_p \cong \mathbb{Z} / p^n \mathbb{Z}$$

□

We will now create the link with previous definitions of the ring $\mathbb{Z}_p$ and the field $\mathbb{Q}_p$. From last proposition we get, $\forall n \geq 1$, a surjective homomorphism

$$\phi_n : \mathbb{Z}_p \longrightarrow \mathbb{Z} / p^n \mathbb{Z}.$$

This family of these homomorphisms induces the homomorphism

$$\mathbb{Z}_p \longrightarrow \varprojlim_n \mathbb{Z} / p^n \mathbb{Z}.$$

Now we can identify both definitions given for $\mathbb{Z}_p$ (and also for $\mathbb{Q}_p$).

(7.21) Proposition. The homomorphism

$$\Phi : \mathbb{Z}_p \longrightarrow \varprojlim_n \mathbb{Z} / p^n \mathbb{Z}$$

is an isomorphism.

Proof: If $x \in \ker(\Phi)$, then $x \in p^n \mathbb{Z}_p$ for all $n \geq 1$ since $x \in \ker(\phi_n)$, i.e., $|x|_p \leq \frac{1}{p^n}$ for all $n \geq 1$, so that $|x|_p = 0$ and thus $x = 0$. This shows injectivity. Now, any element of

$\varprojlim \mathbb{Z}/p^n\mathbb{Z}$ is given by a sequence of partial sums

$$s_n = \sum_{i=0}^{n-1} a_i p^i, \quad 0 \leq a_i < p.$$

By (7.7) this sequence is a Cauchy sequence in $\mathbb{Z}_p$, and hence converges to an element

$$x = \sum_{i=0}^{\infty} a_i p^i \in \mathbb{Z}_p.$$

Since

$$x - s_n = \sum_{i=n}^{\infty} a_i p^i \in p^n \mathbb{Z}_p,$$

we have $x \equiv s_n \pmod{p^n}$ for all n , i.e., x is mapped to an element $(s_1, s_2, s_3, \dots)$ of $\varprojlim \mathbb{Z}/p^n\mathbb{Z}$ which is defined by the given sequence $(s_n)_{n \in \mathbb{N}}$. This proves surjectivity. $\square$

Chapter 8

Valuations

8.1 Definition and Properties

(8.1) Definition . A valuation (multiplicative valuation or absolute value) on a field K is a function

$$|\cdot| : K \rightarrow \mathbb{R}$$

satisfying the properties, $\forall x, y \in K$:

1. $|x| = 0 \Leftrightarrow x = 0$,
2. $|x| \geq 0$,
3. $|xy| = |x||y|$,
4. $|x + y| \leq |x| + |y|$. (Triangle inequality)

Remark. Observe that setting $|x| = 1$ for all $0 \neq x \in K$ and $|0| = 0$, gives us the trivial valuation on K . So, we assume that $|\cdot|$ is non-trivial in this context. Also, defining $d : K \times K \rightarrow \mathbb{R}$ by $d(x, y) = |x - y|$, $x, y \in K$, d gives us a metric on K and a topological structure on K .

(8.2) Lemma . Let $|\cdot|$ be a valuation on K . Then,

1. $|1| = 1$,
2. $|\zeta| = 1$, for all $\zeta \in K$ with $\zeta^d = 1$ for some $0 \neq d \in \mathbb{N}$ (ζ is a root of unity)
3. $|x^{-1}| = |x|^{-1}$,
4. $||x| - |y|| \leq |x - y|$.

Proof: 1. $|1|^2 = |1^2| = |1| \Rightarrow |1| = 1$.

2. $|\zeta|^d = |\zeta^d| = |1| = 1 \Rightarrow |\zeta| = 1$
3. $1 = |xx^{-1}| = |x| |x^{-1}| \Rightarrow |x^{-1}| = |x|^{-1}$,
4. Set $a := x - y, b := y$. Thus, $|a + b| \leq |a| + |b|, \implies |x| - |y| = |a + b| - |b| \leq |a| = |x - y|$.

(8.3) Definition . Two valuations on K are equivalent if they define the same topology on K .

Note: Two metric spaces are equivalent if for any $x \in X$ and any $\delta > 0$, $\exists \epsilon_1, \epsilon_2 > 0$ s.t

$$B_{d_1}(x, \epsilon_1) \subseteq B_{d_2}(x, \delta) \text{ and } B_{d_2}(x, \epsilon_2) \subseteq B_{d_1}(x, \delta)$$

(8.4) Theorem . Let $|\cdot|_1$ and $|\cdot|_2$ be two valuations on K . Then, they are equivalent iff, $\exists$ a real $s \geq 0$ s.t

$$|x|_1 = |x|_2^s, \forall x \in K.$$

Proof: If we have $|x|_1 = |x|_2^s, s > 0$, clearly the two valuations are equivalent since for $\epsilon > 0$,

$$B_{d_1}(x, \epsilon^s) \subseteq B_{d_2}(x, \epsilon) \text{ and } B_{d_2}(x, \epsilon^{1/s}) \subseteq B_{d_1}(x, \epsilon)$$

For converse, if $|\cdot|_1$ and $|\cdot|_2$ are equivalent, a series converging to 0 w.r.t $|\cdot|_1$ will converge to 0 with respect to $|\cdot|_2$ and vice versa. Also, for all K field and $|\cdot|$ on K , the inequality $|x| < 1$ is equivalent to saying that the sequence $\{x^n\}_{n \in \mathbb{N}}$ converges to 0. Therefore, we have

$$|\cdot|_1 \text{ and } |\cdot|_2 \text{ are equivalent} \implies$$

$$(x^n \rightarrow 0 \text{ w.r.t } |\cdot|_1 \iff x^n \rightarrow 0 \text{ w.r.t } |\cdot|_2) \iff (|x|_1 < 1 \iff |x|_2 < 1).$$

Take $y \in K$ an element s.t $|y|_1 > 1$ and let $x \in K, x \neq 0$. Then there exists $\alpha \in \mathbb{R}$ such that $|x|_1 = |y|_1^\alpha$. Let $\left\{\frac{m_i}{n_i}\right\}_{i \in \mathbb{N}}$ be a sequence of rational numbers ($m_i \in \mathbb{Z}, n_i \in \mathbb{N}$)

that converges to α from above. Then $|x|_1 = |y|_1^\alpha < |y|_1^{\frac{m_i}{n_i}}$ as a^x for $a > 1$ is increasing function and hence

$$\left| \frac{x^{n_i}}{y^{m_i}} \right|_1 < 1 \implies \left| \frac{x^{n_i}}{y^{m_i}} \right|_2 < 1$$

Hence, $|x|_2 < |y|_2^{\frac{m_i}{n_i}} \implies |x|_2 \leq |y|_2^\alpha$. Now, take a sequence $\left\{\frac{m_i}{n_i}\right\}_{i \in \mathbb{N}}$ that converges to α from below. It gives $|x|_2 \geq |y|_2^\alpha$ and therefore $|x|_2 = |y|_2^\alpha$. So, for all $0 \neq x \in K$, we have

$$\frac{\log |x|_1}{\log |x|_2} = \frac{\log |y|_1}{\log |y|_2} =: s$$

and thus $|x|_1 = |x|_2^s$. Also, $|y|_1 > 1 \implies |y|_2 > 1$ from equivalence and so $s > 0$.

(8.5) Corollary . Two valuations $|\cdot|_1$ and $|\cdot|_2$ on K are equivalent iff

$$|x|_1 < 1 \iff |x|_2 < 1.$$

(8.5) Definition . A valuation is **non-Archimedean** if $|n|$ is bounded, $\forall n \in \mathbb{N}$. Otherwise, called **Archimedean**

(8.6) Proposition . The valuation $|\cdot|$ is nonarchimedean iff it satisfies the given strong triangle inequality

$$|x + y| \leq \max\{|x|, |y|\}.$$

Proof: If this inequality holds, then we have

$$|n| = |1 + \dots + 1| \leq 1.$$

For converse, let $|n| \leq N \forall n \in \mathbb{N}$. Let $x, y \in K$ and suppose $|x| \geq |y|$. Then $|x|^\nu |y|^{n-\nu} \leq |x|^n$ for $\nu \geq 0$ and we get

$$|x + y|^n \leq \sum_{\nu=0}^n \left| \binom{n}{\nu} \right| |x|^\nu |y|^{n-\nu} \leq N(n+1) |x|^n,$$

thus

$$|x + y| \leq N^{1/n} (1 + n)^{1/n} |x| = N^{1/n} (1 + n)^{1/n} \max\{|x|, |y|\},$$

and hence $|x + y| \leq \max\{|x|, |y|\}$ by letting $n \rightarrow \infty$. □

Remark: Observe that if $|\cdot|$ is non-Archimedean, then for $x, y \in K$,

$$|x| \neq |y| \implies |x + y| = \max\{|x|, |y|\}.$$

Proof: w.l.o.g Assume $|x| > |y|$, then directly $|x + y| \leq \max\{|x|, |y|\} = |x|$. On another side, $|x| = |x - y + y| \leq \max\{|x - y|, |y|\}$. We assume that $\max\{|x + y|, |y|\} = |y|$, but $|x| \leq |y| < |x|$. Contradicting the hypothesis $|x| > |y|$, hence $\max\{|x + y|, |y|\} = |x + y|$ that implies $|x| \leq |x + y|$.

8.2 Ostrowski's Theorem

(8.7) Theorem . Every valuation on $\mathbb{Q}$ is equivalent to the valuations $|\cdot|_p$ or $|\cdot|_\infty$.

Proof. Let $\|\cdot\|$ be a non-Archimedean valuation on $\mathbb{Q}$. Then,

$$\|n\| = \|1 + \cdots + 1\| \leq \max\{\|1\|, \dots, \|1\|\} = 1$$

There also exists a prime $p > 1$ such that $\|p\| < 1$. If not, then all $\|p\| = 1$ and so by uniform prime factorization $\|\cdot\|$ is the trivial valuation. Now, consider the set

$$A = \{a \in \mathbb{Z} \mid \|a\| < 1\}.$$

Here, A is an ideal of $\mathbb{Z}$ (For, $a, b \in A$ and $r \in \mathbb{Q}$, $\|a + b\| \leq \max\{\|a\|, \|b\|\} < 1$, and $\|ra\| = \|r\|\|a\| < 1$). Also $p\mathbb{Z} \subseteq A \neq \mathbb{Z}$ ($\|1\| = 1$ and $1 \in \mathbb{Z}$). Since $p\mathbb{Z}$ is a maximal ideal, we get that $A = p\mathbb{Z}$. Take $a \in \mathbb{Z}$, $a = p^m b$ with $p \nmid b$. Then $p \nmid b \implies b \notin A$ and thus $\|b\| = 1$. This provides

$$\|a\| = \|p^m b\| = \|p\|^m = |a|_p^s,$$

with $s := -\frac{\log \|p\|}{\log p}$. Hence, $\|\cdot\|$ is equivalent to $|\cdot|_p$. Now assume that $\|\cdot\|$ is Archimedean.

Claim: For all integer $m, n > 1$, we have

$$\|m\|^{\frac{1}{\log m}} = \|n\|^{\frac{1}{\log n}}.$$

Proof. we write

$$m = a_0 + a_1 n + \cdots + a_r n^r$$

with $r, a_i \in \mathbb{N}, 0 \leq a_i < n, 0 \leq i \leq r$ and $n^r \leq m$, $\implies r \leq \frac{\log m}{\log n}$. Also,

$$\|a_i\| \leq \|1 + \dots + 1\| \leq a_i \|1\| \leq n$$

gives us that

$$\|m\| \leq \sum_{i=0}^r \|a_i\| \|n\|^i \leq \sum_{i=0}^r \|a_i\| \|n\|^r \leq \left(1 + \frac{\log m}{\log n}\right) n \|n\|^{\frac{\log m}{\log n}}$$

Replace m by $m^k, k \in \mathbb{N}$, and take the k -th root, then we obtain that

$$\|m\| = \sqrt[k]{\|m^k\|} \leq \sqrt[k]{\left(1 + \frac{k \log m}{\log n}\right) n \|n\|^{\frac{k \log m}{\log n}}} = \|n\|^{\frac{\log m}{\log n}} \sqrt[k]{\left(1 + \frac{k \log m}{\log n}\right) n}$$

So when $k \rightarrow \infty$ we have

$$\|m\| \leq \|n\|^{\frac{\log m}{\log n}} \text{ and so } \|m\|^{\frac{1}{\log m}} \leq \|n\|^{\frac{1}{\log n}}.$$

Reversing the roles of m and n we get the second inequality. Since $\|n\|^{\frac{1}{\log n}} > 0, \exists s \in \mathbb{R}$ s.t $e^s = \|n\|^{\frac{1}{\log n}} \implies \|n\| = e^{s \log n}$. Therefore, $\forall x \in \mathbb{Q}, x > 0$, we have

$$\|x\| = e^{s \log x} = x^s = |x|^s$$

As $\|x\| = \|-x\|$, we have that $\|\cdot\|$ is equivalent to $|\cdot|_\infty$. $\square$

We use the symbol ∞ with the usual convention.

8.3 Exponential Valuation

(8.8) Definition. An exponential valuation on K is a function

$$v : K \rightarrow \mathbb{R} \cup \{\infty\}$$

with the properties, $\forall x, y \in K$:

1. $v(x) = \infty \Leftrightarrow x = 0$,
2. $v(xy) = v(x) + v(y)$,

3. $v(x + y) \geq \min\{v(x), v(y)\}$.

Observe that setting $v(x) = 0 \forall 0 \neq x \in K$ and $v(0) = \infty$, we have a trivial valuation on K . We assume that v is non-trivial in this context.

(8.9) Lemma . Let v be valuation on K . Then,

1. $v(1) = 0$,
2. $v(\zeta) = 0$, for all $\zeta \in K$ root of unity,
3. $v(x^{-1}) = -v(x), \forall x \in K^*$,
4. if $x, y \in K$ and $v(x) \neq v(y)$, $v(x + y) = \min\{v(x), v(y)\}$.

Proof: 1. $v(1) = v(1^2) = v(1) + v(1) \Rightarrow v(1) = 0$.

2. Let $\zeta \in K$ with $\zeta^d = 1$ for some $0 \neq d \in \mathbb{N}$. We have $dv(\zeta) = v(\zeta^d) = v(1) = 0$, thus $v(\zeta) = 0$.

3. $0 = v(1) = v(xx^{-1}) = v(x) + v(x^{-1}) \Rightarrow v(x^{-1}) = -v(x)$.

4. W.l.o.g. assume that $v(x) > v(y)$ and so $y \neq 0$. If $x = 0$, then trivial. Assume $x \neq 0$. Then $v(x + y) \geq \min\{v(x), v(y)\} = v(y)$. We have also $v(y) = v(y + x - x) \geq \min\{v(x + y), v(x)\}$. Assume that $\min\{v(x + y), v(x)\} = v(x)$, then $v(x) > v(y) \geq v(x)$ and therefore getting contradiction. So, $\min\{v(x + y), v(x)\} = v(x + y)$ and finally $v(y) \geq v(x + y)$.

(8.10) Definition . An exponential valuation v on K is called **discrete** if $v(K^*) = s\mathbb{Z}$, for a real $s > 0$. And, v is **normalized** if $s = 1$.

(8.11) Definition . Two exponential valuations v_1 and v_2 on K are equivalent if there exists a real $s > 0$ such that $v_1 = sv_2$.

Observe that if we have a discrete valuation on K with $v(K^*) = s\mathbb{Z}$, dividing it by s we obtain an equivalent normalized valuation.

(8.12) Theorem . Let $|\cdot|$ be a non-Archimedean valuation on K and $s \in \mathbb{R}, s > 0$, then the function

$$v_s : K \longrightarrow \mathbb{R} \cup \{\infty\}$$

$$x \mapsto \begin{cases} -s \log |x| & \text{if } x \neq 0 \\ \infty & \text{if } x = 0 \end{cases}$$

is a valuation on K . Moreover, if $s, s' \in \mathbb{R}, s, s' > 0$ and $s \neq s'$, v_s is equivalent to $v_{s'}$. Conversely, if v is an exponential valuation on K and $q \in \mathbb{R}, q > 1$, the function

$$| \cdot |_q : \longrightarrow \mathbb{R}$$

$$x \longrightarrow \begin{cases} q^{-v(x)} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is a non-Archimedean valuation on K . Besides, if $q, q' \in \mathbb{R}, q, q' > 1$ and $q \neq q'$, $| \cdot |_q$ is equivalent to $| \cdot |_{q'}$.

Proof: We need to only check the definitions of valuation and exponential valuation. We start with v_s . Clearly we have $v_s(x) = \infty$ iff, $x = 0$. Let $x, y \in K$, if $x = 0$ or $y = 0$, $xy = 0$ and $\infty = v_s(xy) = v(x) + v(y) = \infty$. Assume $x, y \neq 0$, then

$$v_s(xy) = -s \log |xy| = -s \log(|x||y|) = -s \log |x| - s \log |y| = v_s(x) + v_s(y).$$

If $x = y = 0$, then $\infty = v_s(x + y) = \min \{v_s(x), v_s(y)\} = \infty$. If $x = 0, y \neq 0$ (or $y = 0, x \neq 0$), $v_s(x + y) = v_s(y) = \min \{v_s(x), v_s(y)\}$ (or $v_s(x + y) = v_s(x) = \min \{v_s(x), v_s(y)\}$). Assume now $x, y \neq 0$. We have

$$\begin{aligned} v_s(x + y) &= -s \log |x + y| \\ &\geq -s \log(\max\{|x|, |y|\}) \\ &= \min\{-s \log |x|, -s \log |y|\} \\ &= \min\{v_s(x), v_s(y)\} \end{aligned}$$

Hence v_s is a valuation on K . Assume now $s, s' > 0, s \neq s'$. For all $0 \neq x \in K$, we have

$$v_s(x) = -s \log |x| = \left(\frac{s}{s'}\right) (-s' \log |x|) = \frac{s}{s'} v_{s'}(x).$$

This implies that v_s and $v_{s'}$ are equivalent. Now, we continue with $| \cdot |_q$ for the second case. Need to prove that it is a valuation. Clearly we have that $|x| = 0$ iff, $x = 0$ and since $q > 1 > 0$ that $|x|_q \geq 0$ for all $x \in K$. Let $x, y \in K$, if $x = 0$ or $y = 0$, $xy = 0$ and $0 = |xy|_q = |x|_q |y|_q = 0$. Assume $x, y \neq 0$, then

$$|xy|_q = q^{-v(xy)} = q^{-v(x)-v(y)} = q^{-v(x)} q^{-v(y)} = |x|_q |y|_q.$$

Since $\max\{|x|_q, |y|_q\} \leq |x|_q + |y|_q$, it is enough to show that $|x + y|_q \leq \max\{|x|_q, |y|_q\}$. Hence, if $x = y = 0$, $0 = |x + y|_q = \max\{|x|_q, |y|_q\} = 0$. If $x = 0, y \neq 0$ (or $y = 0, x \neq 0$), $|x + y|_q = |y|_q = \max\{|x|_q, |y|_q\}$ (or $|x + y|_q = |x|_q = \max\{|x|_q, |y|_q\}$). Assume now $x, y \neq 0$. We have

$$\begin{aligned} |x + y|_q &= q^{-v(x+y)} \\ &\leq q^{-\min\{v(x), v(y)\}} \\ &= \max\{q^{-v(x)}, q^{-v(y)}\} \\ &= \max\{|x|_q, |y|_q\} \end{aligned}$$

Therefore, $|\cdot|_q$ is a non-Archimedean valuation on K . Now assume $q, q' > 1, q \neq q'$ and set $r := \frac{\log q}{\log q'}$. For all $0 \neq x \in K$, we have

$$|x|_q = q^{-v(x)} = (q')^{-rv(x)} = |x|_{q'}^r$$

Therefore, $|\cdot|_q$ and $|\cdot|_{q'}$ are equivalent. □

(8.13) Theorem . Let K be the given field with v be a valuation on K . Denote $|\cdot|$ the corresponding non-Archimedean valuation. Then:

1. the subset

$$\mathcal{O} := \{x \in K \mid v(x) \geq 0\} = \{x \in K \mid |x| \leq 1\}$$

is a valuation ring of K (i.e ;an integral domain of $K \forall 0 \neq x \in K$, either $x \in \mathcal{O}$ or $x^{-1} \in \mathcal{O}$),

2. the set

$$\mathcal{O}^* := \{x \in K \mid v(x) = 0\} = \{x \in K \mid |x| = 1\}$$

is the group of units of $\mathcal{O}$,

3. with the set

$$\mathfrak{p} := \mathcal{O} \setminus \mathcal{O}^* = \{x \in K \mid v(x) > 0\} = \{x \in K \mid |x| < 1\} = \{x \in \mathcal{O} \mid x^{-1} \notin \mathcal{O}\}$$

is the unique maximal ideal of $\mathcal{O}$, thus a local ring. Moreover, two equivalent valuations on K give the same valuation ring.

Proof: We will work with the exponential valuation v on K . We can easily see that if two exponential valuation v and v' on K are equivalent, i.e., there is a positive real s such that $v = sv'$, then $\mathcal{O}, \mathcal{O}^*$ and $\mathfrak{p}$ are the same since if $v(x) \geq 0$, resp. $v(x) = 0$,

resp. $v(x) > 0$, $\implies v'(x) = s^{-1}v(x) \geq 0$, resp. $v'(x) = s^{-1}v(x) = 0$, resp. $v'(x) = s^{-1}v(x) > 0$.

To prove that $\mathcal{O}$ is a integral domain, it is enough to show that it is closed under addition and multiplication and for every element in $\mathcal{O}$, it has an additive inverse in $\mathcal{O}$, since all the remaining properties are inherited from the field K . Here, $0, 1 \in \mathcal{O}$ since $v(0) = \infty > v(1) = 0 \geq 0$. Take $0 \neq x \in \mathcal{O}$, then $-x \in K$. We already have $\infty = v(0) = v(x - x) > \min\{v(x), v(-x)\}$ as $x \neq 0$ (and so $-x \neq 0$). By property 4. of Lemma (8.9), then, $v(-x) = v(x) \geq 0$, hence $-x \in \mathcal{O}$. Let $x, y \in \mathcal{O}$. Then $v(xy) = v(x) + v(y) \geq 0 \implies xy \in \mathcal{O}$. Similarly, $v(x + y) \geq \min\{v(x), v(y)\} \geq 0 \implies x + y \in \mathcal{O}$.

For valuation ring, take now $0 \neq x \in K$. If $v(x) \geq 0, x \in \mathcal{O}$, and if $v(x) < 0, v(x^{-1}) = -v(x) > 0$ and hence $x^{-1} \in \mathcal{O}$.

Now take $x \in \mathcal{O}$ and $x^{-1} \in \mathcal{O}$. If $v(x) > 0, v(x^{-1}) = -v(x) < 0$, therefore $v(x) = 0$. The other inclusion is comes from definition. This shows that $\mathcal{O}^*$ is the group of units of $\mathcal{O}$.

Now to prove that $\mathfrak{p}$ is the unique maximal ideal of $\mathcal{O}$. Let $x \in \mathfrak{p}$ and $z \in \mathcal{O}$. Then $v(zx) = v(z) + v(x) \geq v(x) > 0$ and thus $zx \in \mathfrak{p}$. Let now $x, y \in \mathfrak{p}$, then $v(x - y) \geq \min\{v(x), v(-y)\}$. If $y = 0, x - y \in \mathfrak{p}$. If $y \neq 0$, we have that $\infty = v(0) = v(y - y) > \min\{v(y), v(-y)\}$ and, as before, $v(y) = v(-y) > 0$ and already $v(x) > 0$. Thus, $v(x - y) > 0$. This shows that $\mathfrak{p}$ is an ideal of $\mathcal{O}$. Assume that $\exists$ an ideal A of $\mathcal{O}$ such that $\mathfrak{p} \subsetneq A$. Then $x \in A \setminus \mathfrak{p} \implies |x| = 1$ and so $x \in \mathcal{O}^* \cap A$ as. Hence, $A = \mathcal{O}$ and $\mathfrak{p}$ is maximal.

Assume now that there is a maximal ideal $B \neq \mathcal{O}$ of $\mathcal{O}$ such that $B \neq \mathfrak{p}$. Then again $B \cap \mathcal{O}^* \neq \phi$, and so $B = \mathcal{O}$. Hence $\mathfrak{p}$ is unique maximal ideal. $\square$

(8.14) Lemma .Some properties of Valuation Rings :

1. The fraction field of V is K .
2. Any subring of K containing V is a valuation ring of K .
3. V is a local ring.
4. V is integrally closed in K .

Proof. 1 .Now a nonzero element α of K can be written as $\alpha/1$ or as $1/\alpha^{-1}$. Hence $K \subseteq f.f(V)$. The other side follows since $f.f(V)$ is the smallest field containing K .

2. Immediate from the definition of a valuation ring.

3. We will show that the set $\mathcal{M} = \{x \in V | x^{-1} \notin V\}$ of nonunits of V is an ideal. If a

and b are nonzero nonunits in V , then either a/b or $(a/b)^{-1} = b/a \in V$. If $a/b \in V$, then $a + b = b(1 + a/b) \in \mathcal{M}$ (because if $b(1 + a/b)$ were a unit in V , then b would be also a unit). Similarly, if $b/a \in V$, then $a + b \in \mathcal{M}$. If $r \in V$ and $a \in \mathcal{M}$, then $ra \in \mathcal{M}$, otherwise a would become a unit. Hence $\mathcal{M}$ is an ideal and as an ideal of nonunits it is a unique maximal ideal. Hence V is a local ring.

4. Let α be a nonzero element of K , with α integral over V . Then there is an equation like

$$\alpha^n + c_{n-1}\alpha^{n-1} + \cdots + c_1\alpha + c_0 = 0$$

with the c_i in V . If $\alpha \in V$, then we are done. If not, then $\alpha^{-1} \in V$, and if we multiply the above equation of integral dependence by $\alpha^{-(n-1)}$, we get

$$\alpha = -c_{n-1} - c_{n-2}\alpha^{-1} - \cdots - c_1\alpha^{-(n-2)} - c_0\alpha^{-(n-1)} \in V.$$

□

(8.17) Definition . The field $\mathcal{K} := \mathcal{O}/\mathfrak{p}$ is called the residue field of $\mathcal{O}$.

(8.18) Lemma . Let v be a normalized exponential valuation on K , then $\forall x \in K^*$ we have a unique representation $x = ut^n$, where $t \in \mathfrak{p}$ (or t is a prime element) with $v(t) = 1$, $u \in \mathcal{O}^*$ and $n \in \mathbb{Z}$. This t is also called **uniformizer** of $\mathcal{O}$.

Proof: Since $v(K^*) = \mathbb{Z}$ (from normalized definition), $\exists t \in K$ with $v(t) = 1$. Hence $t \in \mathfrak{p}$. Let $0 \neq x \in K$. Then we have that $v(x) = m$ for some $m \in \mathbb{Z}$. Hence $v(xt^{-m}) = 0$ and so $u := xt^{-m} \in \mathcal{O}^*$ and concluding $x = ut^m$.

(8.19) Theorem . Suppose v is a normalized exponential valuation on K , then $\mathcal{O}$ is a principal ideal domain. Hence $\mathcal{O}$ is a discrete valuation ring. Also, all the nonzero ideals of $\mathcal{O}$ are

$$\mathfrak{p}^n = t^n \mathcal{O} = \{x \in K \mid v(x) \geq n\}, \quad n \geq 0,$$

where $t \in \mathfrak{p}$, $v(t) = 1$. Moreover, we have

$$\mathfrak{p}^n / \mathfrak{p}^{n+1} \cong \mathcal{K}, \quad n \geq 1,$$

Proof: Let $\{0\} \neq I \subseteq \mathcal{O}$ be an ideal of the valuation ring and $0 \neq x \in I$ such that $n := v(x) \leq v(y)$ for all $y \in I$. By last lemma, $x = ut^n$ for some prime element

$t \in \mathfrak{p}$ and some $u \in \mathcal{O}^*$. This implies $\mathcal{O}x \subseteq t^n \mathcal{O} \subseteq I$. Now take $y \in I$. Again by last lemma , $y = wt^m, w \in \mathcal{O}^*$. Since $y \in I, m = v(y) \geq n = v(x)$, so we can write $y = (wt^{m-n}) t^n \in t^n \mathcal{O}$, hence $I \subseteq t^n \mathcal{O}$. So $I = t^n \mathcal{O}$ Now consider the homomorphism

$$\begin{aligned} \phi: \mathfrak{p}^n &\longrightarrow \mathcal{O}/\mathfrak{p} \\ at^n &\mapsto a \pmod{\mathfrak{p}} \end{aligned}$$

with $t \in \mathfrak{p}$ a prime element and $a \in \mathcal{O}$. Clearly, ϕ is surjective .Now $\ker(\phi) = 0 \pmod{\mathfrak{p}} \implies \phi(at^n) = \bar{0} \implies a \in \mathfrak{p} \implies \ker(\phi) \subseteq \mathfrak{p}^{n+1}$. Now take $x \in \mathfrak{p}^{n+1} \implies x = at^{n+1} \implies \phi(x) = at \pmod{\mathfrak{p}} = \bar{0}$. Hence $\ker(\phi) = \mathfrak{p}^{n+1}$ Hence,

$$\mathfrak{p}^n/\mathfrak{p}^{n+1} \cong \mathcal{O}/\mathfrak{p}, \quad n \geq 1,$$

□

Chapter 9

Completions

9.1 Cauchy Sequences

(9.1) Definition . Let $(K, |\cdot|)$ be a valued field where $|\cdot|$ is a valuation on K . A sequence $\{a_n\}_{n \in \mathbb{N}}$ in K is called a Cauchy sequence if, $\forall \varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|a_n - a_m| < \varepsilon \quad n, m \geq N$$

(9.2) Definition . Let $(K, |\cdot|)$ be a valued field. Then K is called complete if every Cauchy sequence $\{a_n\}_{n \in \mathbb{N}}$ in K , i.e.

$$\lim_{n \rightarrow \infty} |a_n - a| = 0 \quad \text{w.r.t } |\cdot|$$

(9.3) Lemma . Let K be a complete field with $|\cdot|$ a non-Archimedean valuation on K . Then, for $\{a_n\}_{n \in \mathbb{N}} \subset K$, :

1. the sequence $\{a_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence iff, $\lim_{n \rightarrow \infty} (a_{n+1} - a_n) = 0$,
2. the series $\sum_{n=0}^{\infty} a_n$ converges iff, $\lim_{n \rightarrow \infty} a_n = 0$,
3. Suppose that $\lim_{n \rightarrow \infty} a_n = a \neq 0$, then there exists a positive integer N such that $\forall m \geq N, |a_m| = |a_N| = |a|$.

Proof:1. Suppose $\{a_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence, then from definition, we can have $\forall n \geq N, |a_{n+1} - a_n| < \varepsilon$. For converse, we have $\forall \varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t $\forall$

$n \geq N, |a_{n+1} - a_n| < \varepsilon$. Then, $\forall r, s \geq N$,

$$\begin{aligned} |a_r - a_s| &= \left| \sum_{i=n}^{m-1} (a_{i+1} - a_i) \right| \\ &\leq \max_{n \leq i < m} \{|a_{i+1} - a_i|\} < \varepsilon. \end{aligned}$$

Hence, $\{a_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence.

2. Using $|a_n| = \left| \sum_{i=0}^n a_i - \sum_{j=0}^{n-1} a_j \right| = |s_n - s_{n-1}|$ and 1., the equivalence is trivial.

3. Since $\lim_{n \rightarrow \infty} a_n = a \neq 0$, there exists a positive integer n_a such that, $\forall n \geq n_a, |a_n - a| < |a|$. Then, using the remark of (8.6) we get

$$|a_n| = |a_n - a + a| = \max \{|a_n - a|, |a|\} = |a|$$

□

(9.4)Theorem . Let K be a field and $|\cdot|$ be the valuation on K . Then, there exists a unique, up to K -isomorphism, complete field $\widehat{K}$ with an valuation $|\cdot|_{\widehat{K}}$ such that K is embedded in $\widehat{K}$ as a dense subfield and the valuation on K is a restriction of the valuation on $\widehat{K}$, i.e., $|x|_{\widehat{K}} = |x|$ if $x \in K$.

Proof: It is analogous to what we have done in CH-8 for $\mathbb{Q}$. If R is the set of all the Cauchy sequences in K w.r.t $|\cdot|$, then R is a ring. If $\mathfrak{m} \subset R$ is the set of all the Cauchy sequences that converge to 0, then $\mathfrak{m}$ is a maximal ideal of R . We define the completion field of $(K, |\cdot|)$ as

$$\widehat{K} := R/\mathfrak{m}.$$

with valuation defined as

$$|a|_{\widehat{K}} := \lim_{n \rightarrow \infty} |a_n|,$$

.Also, we have injection of $K \hookrightarrow \widehat{K}$ by sending $a \in K$ to the equivalence class of the Cauchy sequence $(a, a, a, \dots)$. Hence, $K \subset \widehat{K}$. Moreover,

$$\lim_{n \rightarrow \infty} a_n = a$$

in $\widehat{K}$, therefore K is dense in $\widehat{K}$ and $\widehat{K}$ is complete w.r.t $|\cdot|_{\widehat{K}}$.

Let $\widehat{K}'$ be another complete field, w.r.t an valuation $|\cdot|_{\widehat{K}'}$, such that K is dense in $\widehat{K}'$ and, for all $x \in K, |x|_{\widehat{K}'} = |x|$. Take $a \in \widehat{K}$ and let $\{a_n\}_{n \in \mathbb{N}} \subset K$ be a representative of a .

Then, in $\widehat{K}'$, this Cauchy sequence converges to an element $a' \in \widehat{K}'$, because K is dense in $\widehat{K}'$. Define the function $\sigma : \widehat{K} \rightarrow \widehat{K}'$ by $\sigma(a) := a'$. This σ is a K -isomorphism. Also, $|a|_{\widehat{K}} = |\sigma(a)|_{\widehat{K}'}$ because

$$|a|_{\widehat{K}} = \lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} |a_n|_{\widehat{K}'} = |a|_{\widehat{K}'}.$$

9.2 Big Ostrowski Theorem

(9.5) Theorem . Suppose K is a complete field with respect to an archimedean valuation $|\cdot|$. Then K is isomorphic to $\mathbb{R}$ or $\mathbb{C}$ with its archimedean norm equivalent to the usual absolute value of $\mathbb{R}$ or $\mathbb{C}$.

Proof: Since K has archimedean valuation, it has characteristic 0 ; i.e $\mathbb{Q} \subseteq K$. Now the restriction of $|\cdot|$ to $\mathbb{Q}$ must be equivalent to standard absolute value by (8.7). So we may replace $|\cdot|$ with an equivalent absolute value whose restriction to $\mathbb{Q}$ give us the standard absolute value. Now completion of $\mathbb{Q}$ under standard absolute value is $\mathbb{R}$ which is also the closure of $\mathbb{Q}$ inside K .

Hence, we assume w.l.o.g that $(\mathbb{R}, |\cdot|) \subseteq (K, |\cdot|)$ where valuation of K is extension of the usual absolute value of $\mathbb{R}$. To prove that $K = \mathbb{R}$ or $= \mathbb{C}$ we show that each $\xi \in K$ satisfies a quadratic equation over $\mathbb{R}$. As then K will be an algebraic extension of $\mathbb{R}$ and $\mathbb{C}$ is already algebraic closure of $\mathbb{R}$. So let, consider the continuous function $f : \mathbb{C} \rightarrow \mathbb{R}$ defined by

$$f(z) = |\xi^2 - (z + \bar{z})\xi + z\bar{z}|.$$

Observe here that $z + \bar{z}, z\bar{z} \in \mathbb{R} \subseteq K$ and $\lim_{z \rightarrow \infty} f(z) = \infty$. Hence, $f(z)$ has a minimum m (as $\exists M$ s.t $f(z) > f(0) \forall |z| > M$ and $\exists z' \in B_M(0)$ s.t $|f(z')| \leq f(z) \forall z \in B_M(0) \implies f(z') = m \leq f(z) \forall z$). Therefore the set

$$S = \{z \in \mathbb{C} \mid f(z) = m\}$$

is nonempty (as z' exist), bounded (if not then there would be a sequence in S s.t $(x_n) \rightarrow \infty \implies \lim_{n \rightarrow \infty} f(x_n) = m$ ($\rightarrow \leftarrow$)), and closed ($S = f^{-1}(m)$), and there is a $z_0 \in S$ such that $|z_0| \geq |z| \forall z \in S$ (as S is). It is enough to show that $m = 0$, as then we can get the equation $\xi^2 - (z_0 + \bar{z}_0)\xi + z_0\bar{z}_0 = 0$. Assume $m > 0$. Consider the real polynomial

$$g(x) = x^2 - (z_0 + \bar{z}_0)x + z_0\bar{z}_0 + \varepsilon,$$

where $0 < \varepsilon < m$, with the roots $z_1, \bar{z}_1 \in \mathbb{C}$. We have $z_1 \bar{z}_1 = z_0 \bar{z}_0 + \varepsilon$, therefore $|z_1| > |z_0|$ and so

$$f(z_1) > m.$$

For fixed $n \in \mathbb{N}$, consider this real polynomial

$$G(x) = [g(x) - \varepsilon]^n - (-\varepsilon)^n = \prod_{i=1}^{2n} (x - \alpha_i) = \prod_{i=1}^{2n} (x - \bar{\alpha}_i)$$

with roots $\alpha_1, \dots, \alpha_{2n} \in \mathbb{C}$. It follows that $G(z_1) = 0$; say, $z_1 = \alpha_1$. We can substitute $\xi \in K$ into the polynomial

$$G(x)^2 = \prod_{i=1}^{2n} (x^2 - (\alpha_i + \bar{\alpha}_i)x + \alpha_i \bar{\alpha}_i)$$

to get

$$|G(\xi)|^2 = \prod_{i=1}^{2n} f(\alpha_i) \geq f(\alpha_1) m^{2n-1}.$$

From this and the inequality

$$|G(\xi)| \leq |\xi^2 - (z_0 + \bar{z}_0)\xi + z_0 \bar{z}_0|^n + |-\varepsilon|^n = f(z_0)^n + \varepsilon^n = m^n + \varepsilon^n,$$

it implies that $f(\alpha_1) m^{2n-1} \leq (m^n + \varepsilon^n)^2$ and so

$$\frac{f(\alpha_1)}{m} \leq \left(1 + \left(\frac{\varepsilon}{m}\right)^n\right)^2.$$

For $n \rightarrow \infty$ we have $f(\alpha_1) \leq m$, which contradicts the inequality $f(\alpha_1) > m$ that we have proved before. $\square$

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